the magazine for
the contemporary teacher

# TEACHER PLUS

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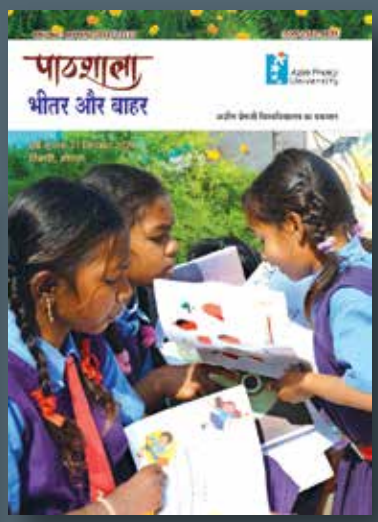


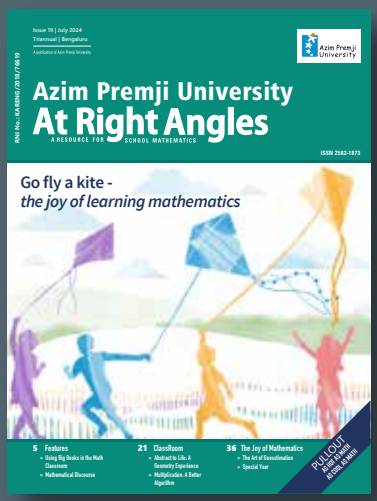

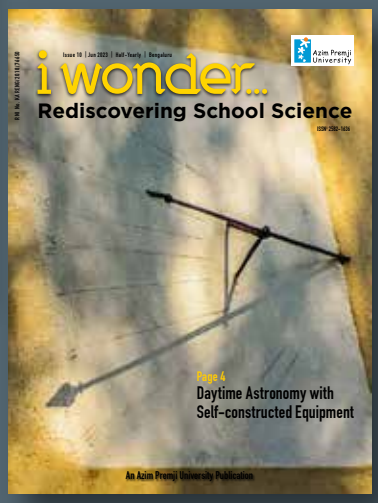

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STEM subjects are very important to solving real-world problems and improving human lives. Why then do we not have as many women as men working in the field of science and technology? Why don't we see more girls choosing the sciences at the higher education level? We try and answer these questions using journal articles and existing data to contextualize the problem of the gender gap in STEM, both globally and specifically in India. We dig into research to see what lies behind this gender gap, and what beliefs, classroom/pedagogical practices, and systemic problems cause this. All is not bleak though. The 13 women scientists we profile here are proof that women are slowly making their mark in these fields. So, how do we as parents and teachers play our part in solving this 'wicked problem? We provide a few suggestions. 8 - 17

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**Cover: Opening the door to STEM magic**

Designed by **Ramya Ramakrishnan**, who wears many hats, including that of an illustrator, product designer, comic book artist, design director, and teacher. She believes storytelling is the most powerful form of communication and strives to incorporate it into her artwork and teaching. You can find her on https://ramyaramakrishnan.com.

# Books and beyond

For people who love books and reading, this is a happy time of year, being the season of literary festivals and book fairs in many parts of the country. While book fairs allow many small and regional language publishers to showcase their catalogues, literary festivals provide a forum for both seasoned and new authors to interact with audiences. One observer estimates that there are over 400 literary festivals held across India – suggesting this is not a phenomenon that is limited to big cities!

This year, I had the opportunity to visit the Hyderabad Book Fair and also participate in the Hyderabad Literary Festival (HLF), and even as I emerged from both with more books than I have shelf space for, I came away with a sense that all is well with the printed word even in this age of screen-immersion. At both events, I saw parents with young children in tow, teachers shepherding lines of uniform-clad pupils, young people lining up to get books signed by admired authors, and people of all ages lingering over stacks of poetry, fiction, and non-fiction. Between sessions at the HL, I noted children with their noses buried in a book, or taking part in one of the many additional activities that have now become an integral part of the event. These included storytelling, art and craft workshops, and stalls set up by scientific institutions that aimed to stoke curiosity and wonder and just possibly, nudge young people toward a career in science. There was much to stimulate thought and crafts to involve the hands.

We've written before in *Teacher Plus* about the value of engaging with the real world, about seeking and using opportunities to take children out of school and into the field. Granted, it's not always easy to organize and does call for some commitment of resources, but it's almost always worth the effort. For teachers too, such events can stoke new ideas and at the very least, point to new books for use in the classroom or for the library – I'm sure you'll agree that browsing at the stalls beats staring at a printed catalogue anyday!

Our cover story this month takes on the vexing issue of the gender imbalance in STEM (science, technology, engineering and mathematics) education in India. Long-time contributor Simran Luthra examines the relatively low presence of women at higher levels in these fields, even though there has been a steady increase in the number of girls taking science and engineering courses. The notion of the "leaky pipeline" has been a matter of concern globally – why and where do the women drop off and what is the impact of their absence in these fields? Yet, there are women who inspire us with their staying power and dedication, as Divya Chowdary writes in a companion piece. What can we as teachers do to encourage girls to "stay with the [science] programme" as they move up the education ladder? Seeing more women at close quarters occupying leadership roles can be quite inspiring, allowing young girls to imagine the possibility. At HLF for instance, children were able to listen to Sowmya Swaminathan, Chief Scientist at the World Health Organization, and environmentalists like Janaki Lenin – and, even better, pick up their books!

Usha Raman

# School uniforms – the fabric of equality

Anuradha C

Photo: Shalini B.
Courtesy: Sandeepam Vidyalaya, Hyderabad

School uniforms, the ubiquitous fabric of our childhood memories, serve as more than just attire. They embody a philosophy, a silent code that governs the lives of millions of school-going children in India. Love them or hate them, uniforms have become an integral part of our educational narrative, weaving together tales of discipline, equality, and nostalgia.

### Why uniforms?

Uniforms are the great equalizers in the cacophony of school life. In a country as diverse as India, where economic and social disparities often stand stark, the humble school uniform serves as a leveller. It ensures that no child's background is betrayed by their attire, allowing their academic abilities and personalities to shine through unimpeded. At a tender age when self-esteem is still fragile, uniforms shield children from the subtle yet damaging judgments based on their economic status.

They also instil a sense of discipline and belonging. The act of donning a uniform every morning is more than a routine; it's a signal that the day's focus will be on learning and growth. The uniform fosters a sense of identity, binding students together as part of a larger collective – be it the green and white of St. Mary's High School or the navy blue and grey of Little Flower Academy.

### The colour conundrum

But oh, the colours! Who decided that dull greys, browns, and blues were the ideal palette for uniforms? Some children's lives are filled with vibrant energy until they step into those drab khaki shorts or navy pinafores, instantly dimming their spark. It's almost as if the uniform is conspiring with the strict mathematics teacher to drain all cheer out of the school day. And let's not forget the eternal mystery: why do white uniforms exist at all? As if schools expect children to defy the laws of nature and stay spotless through recess, art class, and the occasional lunchtime *samosa* spill.

### The stitching saga

The start of a new school year brings with it the perennial uniform rush. Parents, often harried and juggling a million tasks, descend upon local tailors or designated school shops. Tailors, meanwhile, enter their busiest season, transforming their humble workshops into assembly lines of pleats, pockets, and collars. Specialized tailors, who have memorized every school's unique specifications, are a prized find. These unsung heroes battle impossible deadlines, measuring restless children and listening to parental pleas for "a bit of extra margin" because, "you know how quickly kids grow."

For low-income families, however, this rush can be a source of stress. Uniforms are not just another item on the checklist; they are a recurring expense. Children, with their annoying tendency to outgrow everything overnight, often necessitate mid-year replacements. Hand-me-downs can ease the burden, but even these are not a perfect solution – especially when sizes, schools, or styles don't match.

### Skirts, pinafores, and salwar kameez

Uniforms in India are as diverse as the schools themselves. Girls in convent schools often sport skirts and blouses, while others wear pinafores paired with crisp white shirts. In some regions, salwar kameez is the preferred choice, offering comfort and practicality for long school days. Boys' uniforms typically feature shorts or trousers with shirts, though some schools add a tie to the mix – a formality most kids would gladly forgo. Beyond the standard options, some progressive schools now offer gender-neutral uniforms, a small but significant step towards inclusivity.

The variations reflect India's cultural and climatic diversity. What works in a breezy Chennai summer might not be practical for the chill of Shimla. Yet, regardless of the design, uniforms remain the unifying thread, shaping young minds in classrooms across the country.

### Nostalgia in threads

For alumni visiting their old schools, uniforms often become symbols of nostalgia. The sight of children running across the playground in those familiar colours stirs a wave of memories – of scraped knees, stolen lunchboxes, and hushed classroom whispers. The once-despised colours suddenly seem beautiful, a connection to a time when life was simpler, and the biggest worry was finishing homework before the bell rang.

Grown-ups, now dressed in the muted sophistication of corporate attire or the comfort of weekend casuals, often chuckle at their past selves. Who would have thought that the same uniforms they once wished to discard would one day evoke such fondness?

### In conclusion

School uniforms, with their quirks and significance, are more than just garments. They are woven into the fabric of childhood, shaping identities, fostering equality, and teaching life lessons in their own quiet way. From the frantic stitching sessions at the start of the year to the sentimental musings of alumni, uniforms tell stories that transcend generations. And while their colours might never win awards for creativity, they'll forever hold a special place in our hearts – a simple yet profound reminder of the days spent within the four walls of a classroom, dreaming of the world beyond.

Photo courtesy: Greenwood School, Hyderabad

The author is an IT industry drop-out after several years of slogging and money-making. She is now working freelance as a corporate technical trainer and content writer. She is hoping to channelize her passion for writing into a satisfying experience for herself and a joyous experience for her readers. She can be reached at <anuradhac@gmail.com>.

## The persistent STEM gender gap

# Tackling a 'wicked' human problem

Simran Luthra

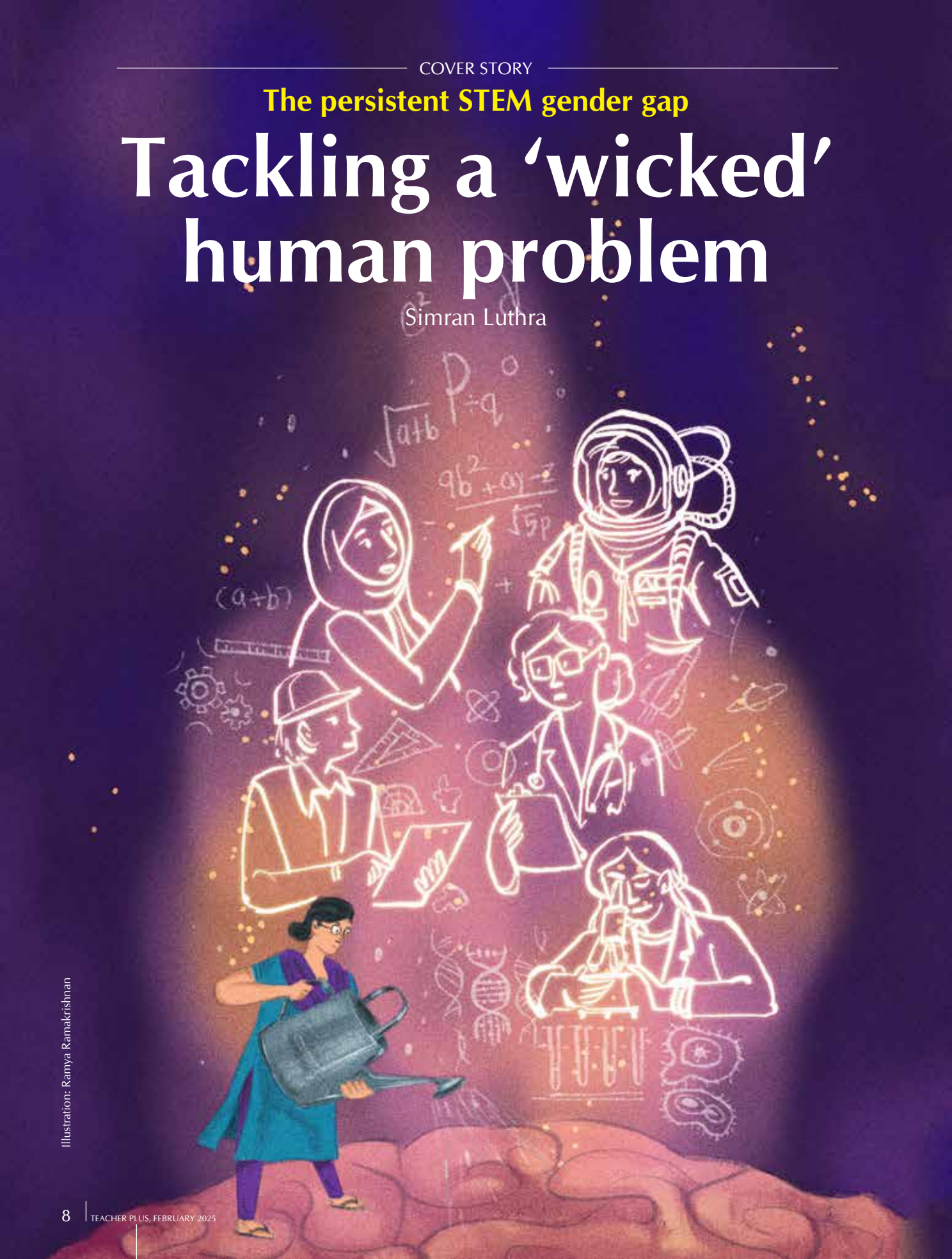

Illustration: Ramya Ramakrishnan

Maths and science subjects have long enjoyed an elevated status among Indian parents. Doing well in these subjects is synonymous with academic excellence and family pride. Even greater pride ensues if children pursue careers in medicine or engineering. This is often at the cost of giving adequate credit to the humanities and social sciences. However, the importance of what is collectively called STEM (Science Technology Engineering Mathematics) is that they do move the needle for humanity in terms of problem-solving real-world problems and innovations that improve human lives.

Despite the importance of STEM, in general and in Indian society in particular, considerable data points to the lack of female participation in STEM in higher education and at the workplace (see References). But first, why does female participation in STEM matter?

### Why is STEM important for women?

Higher participation in STEM is important for girls and women at two levels. First, STEM can help girls and women transition from low-skill and low-paying jobs to higher-paying jobs in terms of careers (studies suggest that women who study science are more likely to get employed and earn up to 28% higher than those who study non-STEM subjects). As per the Digital India Report by McKinsey, around 65 million jobs will be created in India by 2025, and around 40 to 160 million women may need to transition between occupations. In addition, 12 million women are likely to face job losses owing to automation. Hence, STEM education can play a key role in women's social and economic empowerment. However, there is also a larger purpose of advocating for more girls and women in STEM in education and the workforce.

Scientists are ultimately human beings and gender identity plays a crucial role in the kind of problems that scientists choose to solve. It is not surprising then that historically, scientific research and innovations have addressed the perspectives and needs of men. One of the most important examples of this is medical research, which has often excluded women and focused on men. There is, therefore, a huge gap in the understanding of female-specific health issues. Similarly, several technologies have been tested on male users and resulted in precluding the female experience of those same technologies such as safety equipment or workplace ergonomics.

Leadership in the world has been historically dominated by men, including thought leadership. Questions such as, *"What would a world with increased female leadership look like?"* or, *"What would a world with more female participation in STEM look like?"* open up the imagination to newer possibilities and solutions to existing problems or problems never before addressed. Greater involvement of women in STEM could potentially lead to advancements that cater to the needs and problems of girls and women, several of which have been long neglected. Further, more women in STEM research and workplaces would lead to greater inclusiveness and diversity, which in turn can lead to broader perspectives and improved decision-making for various global challenges.

### What does female representation in STEM in India look like?

Before we discuss the gender gap in STEM, it is important to acknowledge the gender gap in education overall. Across the world, women have stood at a disadvantage historically when it comes to education due to systemic barriers such as cultural norms, lack of access to schools, gender biases, and economic constraints, which often prioritized male education over female learning opportunities. Women were not even permitted to attend university till the early 20th century in countries such as the UK. The body of work done by men is therefore much larger than what women have been able to produce. It is hence a game of catch-up for women on various fronts. However, within this already unequal system, women are further underrepresented in STEM in various ways. Following are some of the patterns identified by a few recent studies focusing on India:

- Girls study STEM subjects enthusiastically at the primary school level, but this interest dwindles as they reach adolescence[i].
- One report revealed that a mere 13.5% of STEM faculty members across 98 institutes were female[ii].
- Women make up only 14% of Indian researchers, compared to the global average of 28%[iii].
- Within STEM disciplines also there exists a pattern with engineering having the lowest percentage of women, followed by physics, computer science,

i. https://www.deccanherald.com/education/stem-education-for-girls-the-path-to-equity-2778251
ii. https://timesofindia.indiatimes.com/education/news/stem-in-india-sees-only-13-5-women-faculty-why-is-female-representation-so-low-in-this-field-and-iits/articleshow/109316947.cms#:~:text=A%20BiasWatchIndia%20study%2C%20published%20in,like%20the%20US%20and%20UK
iii. https://genderaveda.cz/en/the-state-of-women-in-science-in-india-not-intelligent-enough/

and chemistry. Mathematics and the earth sciences were next. Biology, however, had the highest rate of female representation primarily because it is a 'softer' science[iv] and aligns with societal expectations of women being caregivers.

- Engineering overall has poor female participation with a mere 6.68% in mechanical engineering and 23% in civil engineering at the undergraduate level.
- Female representation is also significantly low in courses such as Bachelors of Technology (B.Tech) and Masters of Technology (M.Tech)[v].

The situation is not completely bleak, however; gender parity in higher education has seen an increase with the number of female graduates rising from 38.4% in 2014-15 to 42.6% in 2021-22[vi]. Data from the World Bank reveals that India can, in fact, boast of a higher number of female graduates in STEM than other developed countries such as Germany and Australia. However, a large proportion of this population of graduates is also choosing not to join the workforce which renders a huge chunk of human capital underutilized. Further, the larger goals and benefits of having women in STEM remain unrealized.

### Why are women lagging in STEM?

While the STEM gender gap is symptomatic of the larger gender inequities prevalent in society, it is valuable to deconstruct the various factors that have been identified as leading to this gap. A combination of social and structural factors is responsible for such a scenario.

Hence, at a systemic level, in a patriarchal ecosystem, gendered socialization ensures that men are encouraged to pursue STEM subjects while women are not. Therefore, while girls display interest and acumen for science subjects at the primary school level, particularly between the ages of 8 and 10, confidence in their competence for STEM sees a drop by the time they reach adolescence. Various factors play a role in this, including societal stereotypes around boys being better at science and maths than girls are reinforced in various ways.

Parents and teachers are some of the most powerful agents of socialization and often pass on these stereotypes and beliefs in unconscious ways. Hence, the kind of toys parents make available, the sort of activities or experiences they choose for their children, and the kind of advice they give contribute to gendered socialization for boys and girls. Teachers, too, play a significant role in confirming or challenging such societal stereotypes and biases. A prevalent belief is that while girls are more hardworking and bookish, boys have a more natural inclination for science, ask more questions, have better conceptual understanding, and possess a greater ability to come up with original ideas (Subramanian, 2007). Such beliefs influence the pedagogy and language teachers use while teaching, often leading to differential opportunities given to boys and girls. This, in turn, impacts the self-image students have with regard to STEM subjects. Several studies show how girls tend to view subjects such as maths and physics as negatively related to the female gender and that boys too tend to view maths as negatively related to the female gender but maths and physics as being positively related to the male gender (Markova & Herzog, 2015). Such attitudes ultimately ensure that girls are socialized into choosing non-STEM subjects at crucial points such as subject selection during high school and higher education, and the belief that 'science is for men, not women' thus continues to get perpetuated with fewer women taking up STEM.

However, besides gender, other important axes such as caste and class also impact the STEM gender gap. Studies by Sahoo and Klasen (2021[vii]) highlight how despite similar performance levels in maths and science in grade 10 board exams, the gender gap persisted in the choices made by boys and girls. The authors also found that students from socially disadvantaged groups such as Scheduled Castes (SC), Scheduled Tribes (ST), and Other Backward Classes (OBC) had lower rates of choosing science than the general category even when their socio-economic status was not a factor. Hence, it is vital to acknowledge not just gender but the intersection of gender with other social categories such as caste, class, and even location when seeking to understand the gender gap in STEM in India.

iv. https://timesofindia.indiatimes.com/education/news/stem-in-india-sees-only-13-5-women-faculty-why-is-female-representation-so-low-in-this-field-and-iits/articleshow/109316947.cms
v. https://iwwage.org/wp-content/uploads/2024/02/Women-in-STEM-Challenges-and-Opportunities-in-India-Report.pdf
vi. https://www.orfonline.org/expert-speak/women-and-stem-the-inexplicable-gap-between-education-and-workforce-participation
vii. Sahoo, S., & Klasen, S. (2021). Gender Segregation in Education: Evidence from Higher Secondary Stream Choice in India. *Demography, 58*(3), 987-1010. https://www.jstor.org/stable/48687157

For girls who do pursue STEM, the journey is far from easy. Jayasree Subramanian (2007) has explained how gender is invisibilized in science research institutes through the discourses of 'merit' and 'competency'. She argues that although science institutes seem to be completely objective in their stance of giving importance purely to an individual's abilities and talent irrespective of their social background, the fallacy is that they disregard other social realities such as biases inherent in the attitudes of teachers and mentors and the fact that access to quality education itself is not as easy for most in India owing to factors such as caste, class, and location. She also clarifies that in India the gender issue is often reduced to the pressure of marriage and family responsibility, and less attention is paid to the culture and practices within institutes that make it difficult for girls and women to do as well as their male counterparts.

Thus, when women choose STEM and enter higher education, they face various obstacles and challenges. Subramanian (2007) provides various examples of how gender plays out in higher education science institutes in India. For instance, one of the female participants reported having been discouraged by her male professor from applying to the best foreign universities to avoid rejection. She was further discouraged by being told that she would not perform well even if she did get selected. Several PhD scholars pursuing science reportedly face harassment from their supervisors. Taking on women PhD scholars was also considered a risk by male professors as women may get married and not complete the degree. The politics in such institutes were such that women were pushed to do more experimental work than the theoretical work that required more deep thinking. On probing, it was revealed that it was mostly tasks that men disliked doing that were delegated to the women.

Such negative beliefs and behaviour towards women serve as a sort of gatekeeping and discourage women from pursuing science in higher education. Clearly, gendered stereotypes and gendered attitudes exist at all stages and levels, right from early childhood to adulthood, and interventions are required at each of them.

### What are some solutions?

The gender gap in STEM is not a newly discovered problem. It has been acknowledged for decades. Worldwide and in India there are several initiatives aimed at addressing the specific challenges faced by girls and women in STEM. Some India-based initiatives and organizations include the Atal Tinkering Labs, Women for STEM India, Mobile Science Labs, and Vigyan Mitra among others. These initiatives and organizations focus on research, capacity

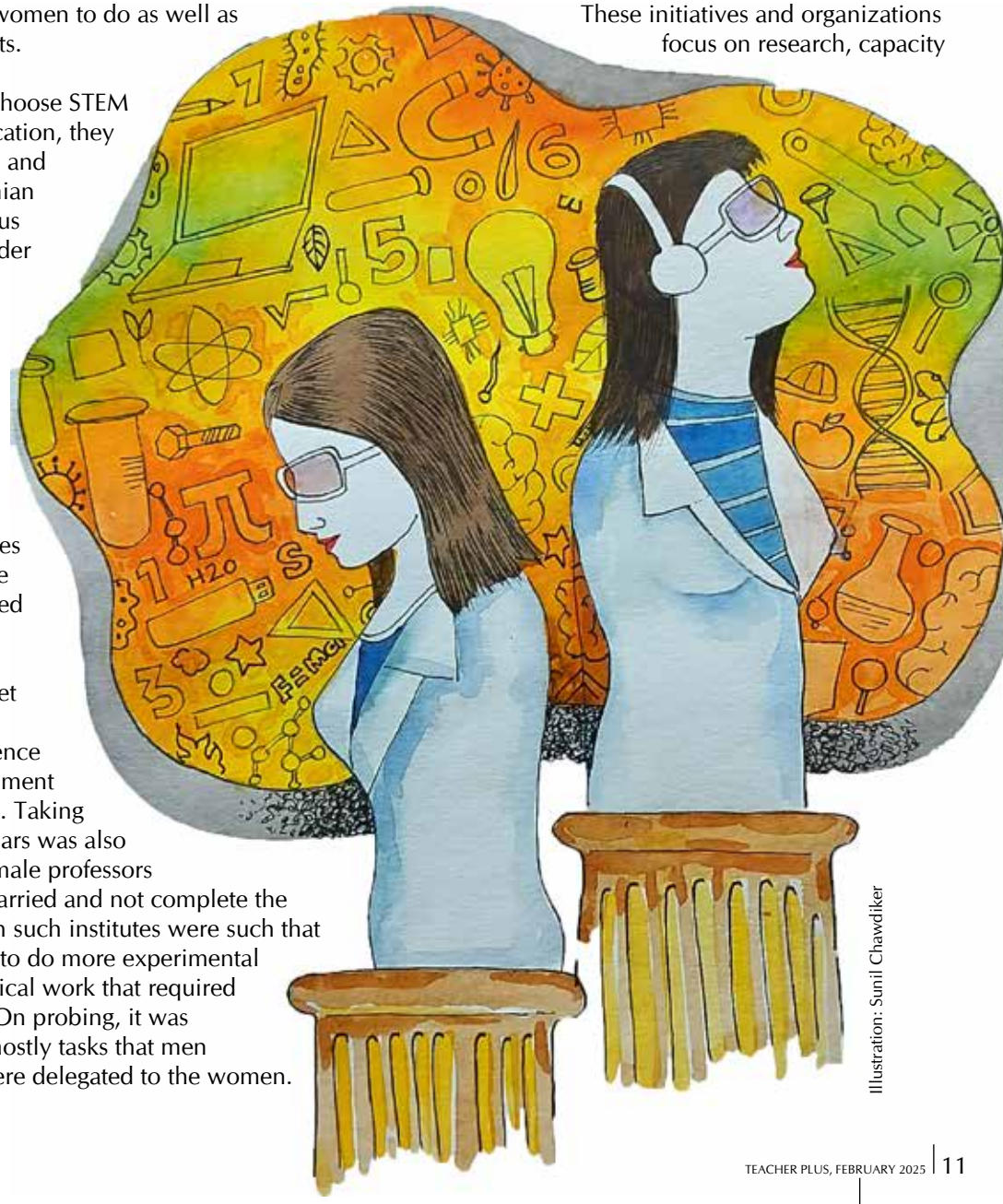


Illustration: Sunil Chawdiker

building, mentorship, and internship opportunities. However, besides such a targeted approach, an overhaul of mindset and attitude towards women is necessary for sustainable, long-term, and real change. This is where teachers, educators, parents, and invested adults can play a pivotal role.

Below are some strategies that teachers and invested adults can practise to create a more gender-equitable environment for children and young adults that can encourage them to pursue their natural interests:

**Practise inclusive pedagogy and providing equal opportunities:** Ample studies (see References) from across the world show how teachers and parents hold gendered beliefs regarding STEM. Such subconscious beliefs get transferred to the younger generations through the attitudes and actions of adults. Thus, a crucial prerequisite for teachers and invested adults is to examine their own biases and beliefs, which could foster stereotypical attitudes that steer girls towards humanities and social sciences and boys towards maths and science.

Teachers should take care that they do not call upon boys routinely to answer certain types of questions or do certain kinds of tasks, and ensure that they are more balanced in the allocation of tasks and activities. Some research points to how the context can also impact the interest levels of students. Kerkhoven et al (2016) point out that the same concept when introduced in a feminine context such as 'discussing the dangers of smoking' was more interesting for girls, than when it was introduced as 'examine which poisons have an effect on the nervous system'. The authors advocate for a mix of feminine and masculine contexts to ensure that different learners are drawn in and engaged with science and mathematics.

Revisiting one's pedagogical style and repertoire from a gender lens to see how well boys and girls respond to different kinds of activities or forms of instruction can be an enriching exercise that teachers engage in for feedback on what changes can be made.

**Carry out a gender audit:** Check if the robotics club in your school is predominantly full of boys. How is the participation of girls in science fairs? Are there more girls in arts and crafts clubs than boys? Instead of letting such things go unnoticed or normalizing them as differences between genders, it is vital to take stock of and remedy such scenarios through a holistic investigation of why this is the case, and then adopting relevant approaches to foster an environment where children can follow their natural inclinations without the burden of gender stereotypes and expectations. It is critical to be wary of surface-level fixes such as trying to increase the number of girls in robotics clubs without truly engaging with why that is the case.

**Use gender-inclusive content:** The importance of non-stereotypical representation of men and women as well as of role models cannot be emphasized more. Children need to see other girls and women engaged in STEM subjects to foster a sense of possibility and belonging. Educational material and resources ranging from textbooks to digital content need to go beyond tokenistic representations of women or stereotypical representations, (such as scientists and engineers as male), and ensure that resources are gender-balanced and the language used is gender-neutral.

Again, such efforts towards gender-neutral and gender-balanced content need to be made at every stage of education right from early childhood to higher education, as such beliefs constantly get formed and reshaped in the minds of young children and adolescents. Hence, if textbooks and other teaching-learning materials are not aligned with such representations, teachers and parents can curate or create supplementary material to ensure that girls and women are not shown in passive roles (care-giving, nursing) but in roles where they are taking the lead, or that the number of times men are depicted is not more than that of women. A gender audit of textbooks and educational material used, including literature readers, can be an important and useful exercise to conduct as an input for the next steps.

**Involve role models and mentors:** Teachers and/or parents can create and build networks with women in STEM and invite them as role models or mentors formally or informally. It can be extremely inspiring for young children to meet and interact with other girls/women working in STEM to truly demonstrate that gender need not be a barrier to success in science, technology, engineering, and math fields. Normalizing women in STEM can be powerful for shaping the future aspirations of girls and their acceptance in STEM. At times, women working in STEM may also be needed to facilitate advocacy that can influence families and communities to understand that STEM is a viable career choice for girls.

**Effect real change in our lives as adults:** Perhaps the most powerful step that educators and invested adults can take is to lead lives that reflect gender equality. Collaboration and respect between genders, checking one's own biases as well as one's privilege regarding gender, caste, and class are essential for setting an example for children and contribute to reducing discrimination of various kinds that plague society. For teachers, this could even mean examining the social dynamics of their staff rooms. Anecdotal evidence suggests that gendered behaviour is common in most Indian staffrooms with women teachers being expected to take on domestic tasks (like caring for sick children and planning important school days) and men teachers being more passive. The division of non-academic tasks also tends to be gendered with men involved in more masculine work such as disciplining or rapport-building with students.

### Let us remember that meaningful change takes time

A 'wicked problem' is one that is elusive and complex with no clear solutions. It often even seems resistant to resolution. Climate change, education, poverty, and gender are some of the biggest and most important wicked problems facing humankind. Collaborative problem-solving, systems thinking, policy innovations, participatory approaches, and shifting mindsets and cultures are all required to make progress with such wicked problems. The STEM gender gap is not a very recent problem. It has been around for a while. A meta-analysis spanning five decades (Miller et al., 2018) has noted that owing to mass media depicting more women in science and improved efforts towards representation of women in science in the USA, children's drawings of scientists are now slightly more diverse and include more women than earlier. However, children still associate scientists with men as they grow older. Change is painfully slow when it comes to wicked problems. One can compare efforts to the dispersal of seeds and hope that some will sprout soon, while others may take longer. Education, however, is an optimistic endeavour, and educators are idealists who need to carry on even in the face of persistent and wicked problems that seemingly do not change.

To borrow the wise words of American thinker Thomas Merton: "Do not depend on the hope of results. You may have to face the fact that your work will be apparently worthless and even achieve no result at all, if not perhaps results opposite to what you expect. As you get used to this idea, you start more and more to concentrate not on the results, but on the value, the rightness, the truth of the work itself."

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# Modern scientists

## 13 women who've changed our world

Divya Choudary

There's a thrill in witnessing a new discovery come to life – an undeniable rush that crosses boundaries of age, nationality, and experience. Science brims with such breakthroughs. Yet, for a long time, women's efforts often flew under the radar. Fortunately, the tide has shifted. In recent years, brilliant women have been recognized for their game-changing contributions in fields ranging from deep space exploration to public health, advanced computing and sustainable agriculture.

In what follows, we spotlight 13 remarkable scientists – some you may already know, others you may be meeting for the first time. Each has contributed to advancing science in unique ways, overcoming challenges, and exploring new frontiers. Together, they represent a diversity of disciplines, nations, and perspectives, showing that innovation can thrive in any corner of the world.

**Defying astrophysical boundaries: Dr. Katie Bouman**

When the very first image of a black hole mesmerized the world in April 2019, Katie Bouman's name shot into the spotlight. An American computer scientist and electrical engineer, Dr. Bouman led the development of key algorithms that pieced together data from telescopes all over the globe as part of the Event Horizon Telescope (EHT) collaboration. The result was an unprecedented visual of a supermassive black hole in the M87 galaxy – an image many once considered impossible to obtain.

Dr. Bouman's work illuminates how innovative science benefits from fresh perspectives and often demands multidisciplinary teamwork: astrophysicists, computer scientists, and engineers joined forces to achieve something extraordinary. Today, she continues her research at Caltech on computational imaging, pushing the boundaries of how we gather and interpret complex data about the cosmos.

**Steering space exploration: Dr. Ritu Karidhal and Muthayya Vanitha**

Nicknamed the "Rocket Woman of India," Dr. Ritu Karidhal is a senior scientist at the Indian Space Research Organisation (ISRO). Known for her earlier work on the Mars Orbiter Mission (Mangalyaan), she took on the role of Mission Director for Chandrayaan-2, launched in July 2019. Dr. Karidhal guided a team tasked with navigating engineering challenges under tight budgets – elevating India's status in global space exploration.

Alongside Dr. Karidhal, Muthayya Vanitha served as Project Director for Chandrayaan-2 – becoming the first woman to lead an interplanetary mission at ISRO. Although the mission's lander faced a setback during its landing attempt, the orbiter continues to send valuable data on the Moon's geography and mineral composition.

Both scientists broke barriers in a traditionally male-dominated field, redefining the narrative for future generations of women engineers and researchers. Their shared leadership underscores how collaboration, mentorship, and mutual respect can drive not only a mission's success, but also a shift in workplace dynamics.

**Transforming public health: Dr. Gagandeep Kang**

In 2019, Dr. Gagandeep Kang became the first Indian woman to be elected a Fellow of the Royal Society – an honour marking her groundbreaking work in microbiology. Her primary focus has been the rotavirus vaccine, which has drastically reduced child mortality in developing nations. Through her leadership at the Translational Health Science and Technology Institute in India, Dr. Kang champions indigenous research and a collaborative approach to global health.

Her impact goes beyond her lab: she mentors young researchers, fosters partnerships between private and public institutions, and emphasizes solutions rooted in local contexts. Dr. Kang's journey is proof that substantial change can arise when research is tailored to real-world needs, particularly in low-resource settings.

**Pioneer of mRNA vaccines: Dr. Katalin Karikó**

Dr. Katalin Karikó's decades-long dedication to mRNA technology remained largely without recognition – until the COVID-19 crisis demanded a rapid vaccine solution. Her groundbreaking work, conducted with Drew Weissman, laid the scientific foundation for some of the world's first authorized COVID-19 vaccines. This earned them the 2023 Nobel Prize in Physiology or Medicine.

Despite facing repeated rejections for funding and academic positions, Dr. Karikó persisted, demonstrating how transformative science often relies on steadfast belief in a hypothesis – even when met with skepticism. By stabilizing mRNA molecules, she enabled them to deliver genetic instructions to human cells, a method now central to modern vaccines.

Dr. Karikó's contributions not only revolutionized vaccine development during the pandemic but also set the stage for future breakthroughs in cancer immunotherapy, HIV treatment, and other complex diseases. Her story is a testament to resilience, innovation, and the world-changing potential of science.

**Driving COVID-19 vaccine research: Dr. Pragya D. Yadav**

During the global scramble to contain COVID-19, Dr. Pragya D. Yadav emerged as a pivotal figure in India's vaccine development efforts. Stationed at the Indian Council of Medical Research-National Institute of Virology, she led critical phases in designing and testing Covaxin – the nation's first indigenously produced coronavirus vaccine.

Despite the pressure of urgent timelines, Dr. Yadav and her team balanced speed with meticulous validation of the vaccine's safety and efficacy. Their dedication helped deliver a viable immunization option not just to Indian citizens, but also to countries with limited access to vaccines. Her work underscores how local breakthroughs can have global impact, particularly when they address shared challenges such as a pandemic.

**Editing the code of life: Dr. Emmanuelle Charpentier and Dr. Jennifer Doudna**

In 2020, Dr. Emmanuelle Charpentier (France) and Dr. Jennifer Doudna (USA) were jointly awarded the Nobel Prize in Chemistry for pioneering the CRISPR-Cas9 gene-editing tool. Their research and collaboration – bridging continents and scientific disciplines – transformed how we approach genetic diseases. Their work revolutionized our approach to treating genetic diseases from sickle cell anemia to inherited blindness.

Both scientists emphasize ethical considerations alongside discovery: CRISPR's power to "cut and paste" DNA compels the global community to consider how and when it should be used. Dr. Charpentier now leads the Max Planck Unit for the Science of Pathogens in Berlin, while Dr. Doudna helms the Innovative Genomics Institute at the University of California, Berkeley. Their story reminds us that when minds meet across borders, transformative science can unfold – along with a responsibility to ensure it's used wisely.

**Transforming data into insight: Dr. Kerrie Mengersen**

Statistics might not capture headlines as often as rocket launches or gene editing, but Dr. Kerrie Mengersen from Australia shows us why it should. ABayesian statistician at Queensland University of Technology, Dr. Mengersen develops intricate models to interpret large datasets – whether mapping endangered ecosystems or assessing public health risks – providing critical insights that guide decision-making and public policy strategies.

Since 2019, Dr. Mengersen has been at the forefront of projects that merge statistical rigor with real-world applications, showcasing the transformative power of data-driven insights. Her work spans diverse fields from protecting jaguar populations in the Amazon to developing Australia's first interactive cancer map, which aids in visualizing and addressing regional disparities in cancer care.

Awarded the Ruby Payne-Scott Medal in 2024, Dr. Mengersen's career illustrates how meticulously analyzed data can guide targeted interventions in ecology, healthcare, and environmental management. Her achievements underscore the potential of mathematics not just to clarify complex phenomena but to tell inspiring stories – provided we ask the right questions and interpret the results responsibly.

**Innovations for accessibility: Hadeel Ayoub**

Hadeel Ayoub's "BrightSign" glove translates sign language into spoken words in real time, bridging communication gaps for people with speech and hearing impairments. Since 2019, Ayoub, a Saudi-based researcher and innovator, has refined this wearable device, integrating software engineering, linguistics, and user-experience design.

Ayoub's invention also aims to reduce the social distance that technology sometimes fosters. By offering a practical, user-friendly solution, Ayoub embodies the principle that empathy can fuel innovation. Her story is a testament to how modest yet impactful ideas can change lives, especially when developed in conversation with the communities they aim to serve.

**Pioneering ethical AI and computer vision: Dr. Fei-Fei Li**

Artificial intelligence can sometimes feel impersonal or mysterious, but Dr. Fei-Fei Li aims to keep the human spirit firmly at AI's core. A Stanford professor and co-director of the Stanford Institute for Human-Centered Artificial Intelligence, Dr. Li has championed the notion that AI should benefit society as a whole. Known for her crucial work in computer vision – training machines to interpret images – Li has broadened her focus to the social consequences of AI.

Since 2019, she has advocated for research that balances rapid technological advancement with principles of fairness, accountability, and human wellbeing. By actively pushing for diversity in tech, she urges industries to address biases that can permeate AI systems. Dr. Li's comprehensive approach ensures that innovation in AI goes hand in hand with ethical responsibility.

**Reinventing agricultural science: Dr. Segenet Kelemu**

Growing up in rural Ethiopia, Dr. Segenet Kelemu saw firsthand how vulnerable small-scale farmers are to plant diseases and pests. Determined to find sustainable solutions, she studied molecular plant pathology in the US and returned to Kenya where she now directs the International Centre of Insect Physiology and Ecology (icipe).

Since 2019, Dr. Kelemu has advanced research on using insects' symbiotic bacteria to bolster crop resilience, thereby reducing the need for harsh pesticides. By merging community-based approaches with cutting-edge research, Dr. Kelemu has transformed local farming methods. Her focus on community-based science has helped local farmers increase yields and protect the environment. Dr. Kelemu's leadership underlines how breakthroughs in agriculture can be both high-tech and locally attuned – making a direct, positive impact on food security.

**Bringing chemistry to life: Dr. Frances Arnold**

Dr. Frances Arnold made history in 2018 as the first American woman to win the Nobel Prize in Chemistry, recognized for her groundbreaking work on "directed evolution." This innovative method that mimics natural selection in the lab involves designing new enzymes – proteins that catalyze chemical reactions – that improve renewable biofuels, create novel pharmaceuticals, and reduce the environmental impact of chemical manufacturing.

Since then, Dr. Arnold has continued her transformative work at the California Institute of Technology, refining techniques that promise greener industrial processes, advanced pharmaceuticals, and more efficient biofuels. Beyond her scientific achievements, she serves as co-chair of the President's Council of Advisors on Science and Technology in the United States. Her work highlights how emulating nature's trial-and-error approach can drive sustainable solutions. It also demonstrates how innovation often springs from reimagining processes already at work in the natural world.

**Signaling broader shifts**

These 13 women aren't just names on a list; they symbolize a broader shift in how we view science and who shapes its future. From decoding the mysteries of black holes to pioneering vaccine technologies and reshaping agricultural practices, they've challenged outdated assumptions about what women can – and should – achieve. Their journeys illustrate that scientific progress depends on diverse perspectives, where collaboration, resilience, and curiosity light the path forward.

As we celebrate the International Day of Women and Girls in Science, let's remember that the achievements of these researchers, engineers, and inventors in helping us understand the world as we know it, also serve to help us reimagine what the world could become.

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# Smartphones and social media

## Balancing challenges and opportunities in schools

Neerja Singh

Smartphones and social media are now everywhere, changing how people communicate and learn. But their widespread use among students has raised concerns. In India, the issue is serious because of the rapid spread of affordable smartphones and internet access. Schools need to find solutions that work in this context.

Smartphones have changed education in India, especially during the pandemic when digital tools became essential for learning. However, this shift also brought new challenges. A survey of 6,229 parents of schoolchildren aged 6 to 16 across 21 states found that most kids used smartphones for leisure rather than education.

Smartphones also affect students beyond academics. Teachers say students get distracted during lessons when notifications and social media pull their focus away from studies. Students often seem to prefer online interactions to real-life connections, leading to loneliness and poor social skills. In India, the focus on academic success often ignores the need to teach healthy digital habits.

The mental health effects of smartphone addiction are clear. Studies, in India and worldwide, link excessive screen time to anxiety, depression, and poor sleep. Teens face additional stress from trying to maintain a perfect online image. Social media platforms, designed to keep users hooked, make this worse. Cyberbullying and unhealthy comparisons are common problems. In schools, ironically enough, teachers struggle to compete with smartphones for students' attention. Social media also spreads school drama, bullying, and scandals, making things harder for everyone.

Some schools in India are trying new ways to deal with these problems. Many now limit smartphone use during school hours. Some ask students to leave their phones in lockers upon arrival. This creates a distraction-free space but ensures access during emergencies. Other schools have set up "digital detox zones." These areas ban devices entirely, encouraging face-to-face interaction. The Delhi government has banned mobile phones in schools. Parents are asked to ensure that their children do not bring phones to school. If students do bring phones, schools must provide safe storage, like lockers. Teachers and staff are also asked not to use phones during teaching activities, whether in classrooms, labs, playgrounds, or libraries.

The case for phone-free schools is getting stronger. Research shows that smartphones hurt focus, learning, and social connections. A study called "Brain Drain: The Mere Presence of One's Own Smartphone Reduces Available Cognitive Capacity" found that even having a phone nearby, without using it, reduces brain performance. France and Australia have already banned phones in many schools. India could benefit from similar policies. Schools that have banned phones entirely often see better attention and engagement. San Mateo High School in California uses lockable pouches for phones. Students there have been found to focus more on lessons and talk to peers instead of scrolling through social media.

Behavioural programs are also helping. Workshops on digital citizenship and cyberbullying prevention are becoming popular. These programs teach students how to use technology safely and responsibly. For instance, some schools have begun to use WhatsApp groups to connect teachers, students, and parents. They share assignments and updates while setting clear rules to prevent misuse.

In this evolving scenario, mental health support becomes crucial. Schools are adding Social and Emotional Learning (SEL) programs to help students manage stress and build resilience. Mindfulness exercises and counselling are also common. Open conversations about mental health and peer support groups further help students feel valued and supported. Parents and communities play a big role here. Parents shape their children's digital habits.

Schools are holding workshops to teach parents how to monitor screen time and set boundaries. The Hyderabad Police's guide on "Digital Parenting" provides parents with tools to handle kids' screen habits better. It highlights the risks of excessive smartphone use, like cyberbullying and addiction. The guide suggests setting clear rules, monitoring usage, and having open conversations about online safety. It also encourages parents to spend quality time with kids to reduce reliance on devices. Practical tips like using parental controls and creating device-free zones are included. Such efforts help create a balanced approach to technology at home.

However, smartphones also offer unique benefits for education. Schools are using apps, online quizzes, and virtual classrooms to make learning more engaging. Platforms like Google Classroom and YouTube help students access information more easily. But schools need clear rules to balance the benefits and drawbacks of technology.

Policies that limit screen time and monitor online activities ensure smartphones are tools, not distractions.

Government policies can support these efforts. Introducing rules for digital use in schools can make solutions more effective. Adding digital literacy and mental health education to the curriculum will help students navigate the online world better. Policymakers, educators, and tech companies can work together to create tools that enhance learning while reducing distractions. Apps that block non-educational content during school hours could be useful.

The goal is not to ban smartphones completely but to teach students to use them wisely. Schools that focus on emotional, social, and intellectual growth are better prepared to address these challenges. Several schools are experimenting with "Tech-Free Days." Students and teachers spend time on storytelling, art, and games. At first, there may be resistance, but it soon becomes the most popular day of the week.

In the dynamic landscape of Indian education, smartphones and social media represent both a challenge and an opportunity. They have the potential to revolutionize learning, bridging gaps in access and fostering creativity, yet they also threaten to erode focus, mental wellbeing, and meaningful human connections. The solution lies not in outright bans but in cultivating a culture of mindful and balanced use. Schools, parents, and policymakers must collaborate to create environments where technology is an enabler rather than a disruptor – integrating digital literacy, promoting emotional resilience, and encouraging authentic interactions. By embracing a holistic approach that values both innovation and human connection, Indian schools can empower students to thrive, not just in the digital age but in life itself.

A unique perspective on the challenges and opportunities of smartphones and social media in Indian schools is to view them as tools for *cultural preservation and innovation*. In a rapidly globalizing world, where traditional practices and regional identities risk being overshadowed, smartphones can act as bridges connecting students to their heritage. Through curated digital platforms, students can explore folk stories, regional art forms, traditional recipes, and local languages. Schools could create initiatives where students document oral histories or local traditions using their devices, fostering a sense of pride in their cultural identity.

Simultaneously, these same tools can empower innovation by teaching students how to adapt ancient wisdom to solve contemporary challenges – whether through sustainable living inspired by traditional methods or community-based solutions rooted in age-old practices. By blending cultural learning with modern technology, Indian schools can position smartphones not as potential distractions, but as powerful instruments for connecting the past with the future, creating a generation of tech-savvy, culturally grounded, and globally aware citizens. This dual-purpose integration of smartphones – preserving the old while innovating for the new – could transform them from controversial gadgets into indispensable assets for holistic education.

The writer is a generational diversity speaker, author of six generational books, and a consultant on working with GenZ; Leading with Social Impact, and Suicide Prevention and Destigmatisation. A millennial in spirit, with the benefit of hindsight, she uses her 37 years in media and education to help leverage generational diversity at work and at home. She can be reached at https://www.linkedin.com/in/neerja-singh/.

# A book that

**What is the book about?**

This question is best answered using two lines from the book:
"Two of mythology's best-known women"
"What do we know of these women themselves"?

**How did I find the book?**

Few, if any, of you who read this will claim to be unfamiliar with either Sita or Helen. Films, television serials, comic books, religious texts, textbooks, stories by elders at home, songs or other sources would have led you to them. How many of you drew parallels between the two? This surely never occurred to me. And after reading and rereading the book, I wondered why it did not! This is where the book scores: originality. Its ability to surprise, to nudge one to think.

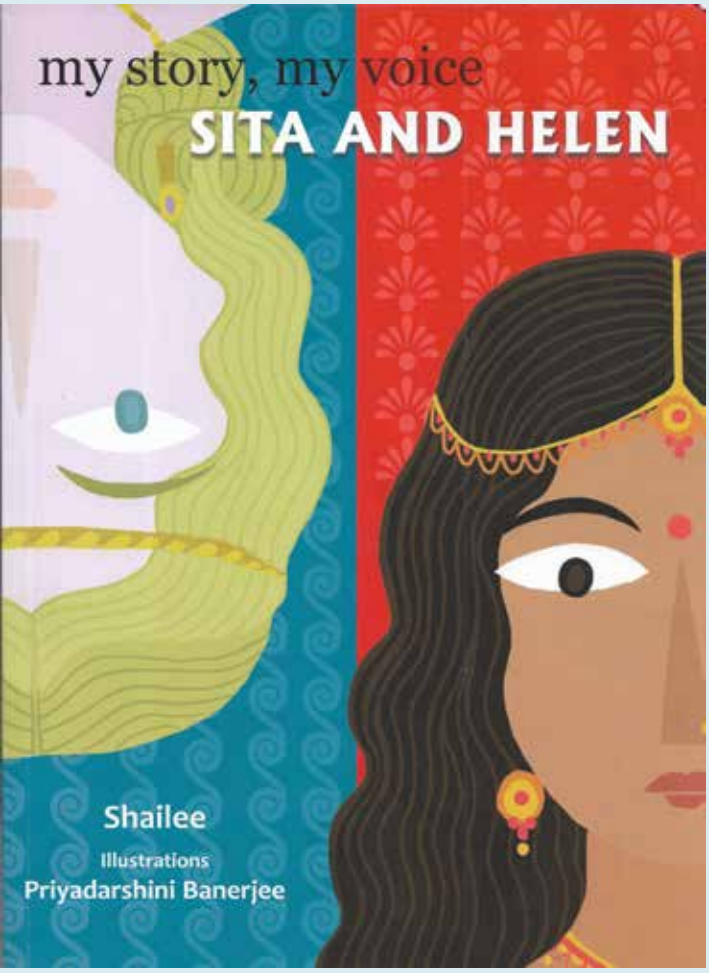
This book takes you through the lives of Sita and Helen though their eyes – a Sita interested in "court happenings" and Helen in "warfare", and a lot else. A Sita and Helen many of us possibly did not know about. This is done at a brisk pace without being text-heavy. Each of the six sections (birth, childhood, wedding, abduction, life in captivity, and the aftermath) talks about both Sita and Helen.

The book underscores the contradictions in their lives as well; the mother of her abductor becomes Helen's "true friend" and Helen "wasn't happy" when her brothers came to "rescue" her. Only one line is common across the two parallel stories. A line which succinctly brings out the crux of the book, a line which would not be difficult to come across today as well, "One day I was informed that I was going to be married."

This is a book where the author and illustrator have taken the story ahead together. Colour schemes for both the characters are different, as are their dresses and accessories. The eyes of the different characters had me glued to the pages! Page numbers are absent; but then is there a need for page numbers in a page-turner like this?

# nudges you to ask questions

Nimesh Ved




**My story, my voice: Sita and Helen**
**Shailee**
Illustrations: Priyadarshini Banerjee
Publisher: Tulika
Price: 345/-

**How can we interpret this book with children and colleagues?**

This is a book that can be perceived as somewhat risky and tricky – especially with children – in today's times, when raising questions about religious beliefs and mythology are considered blasphemous. However, it can also be looked upon as a thought-provoking tool with children of different ages and aptitudes, and of course, with adults too. A book very much welcome.

First of all, this book can have us discuss plurality with children. Ramanujan in his essay *Three Hundred Ramayans* says, "The number of *Ramayanas* and the range of their influence in South and Southeast Asia over the past twenty-five hundred years or more are astonishing." Helen too has been a part of multiple plays and epics including those by Homer, Euripides, and Shakespeare. We can discuss whether stories, like poems, can be interpreted in multiple ways? If characters within a story can be looked upon differently? Is it normal to have opinions and perspectives that are different from others, from the majority? Are our stories, epics open to questions? Do epics, stories change with time?

Can this book encourage us to read these and other epics? To interpret them for ourselves? To explore more about 'Trojan horse', 'Achilles heel' and other idioms? To read?

How do our stories, epics look at the common people, those neither of royal pedigree, nor of elite background? What about the day-to-day lives, family needs, or livelihoods of these people that are disrupted when an entire city is under siege for a decade or is set on fire? Would the epics be possible without them? Are we aware how we, today, perceive the proverbial "man (or woman) on the street"?

How is it that Sita and Helen have so much in common? Are there other themes and elements that are common across our epics? What could be the reasons?

It would also be fun to just let children look at the illustrations and come up with their own interpretations.

The reviewer loves walking, cycling and is often seen amidst books. He enjoys putting his thoughts in words, blogs at https://nimesh-ved.blogspot.com/, and can be reached at <nimesh.explore@gmail.com>.

# Harmony in learning

## Cultivating diversity and inclusion in school communities

Sanjhee Gianchandani

Inclusion in the classroom means that all students, irrespective of their racial, religious, or ethnic background, gender, learning style, or ability have equal access to educational opportunities in a learning environment where they are equally safe, valued, and respected. Inclusion in the classroom is immensely important because it fosters each child's sense of belonging, the feeling of being seen and understood for who they are, and their inherent value as a human being. Inclusive classrooms empower children to grow and thrive as self-assured individuals, not just as students, but in life as citizens of the world. Within inclusive school cultures, educators deal with fewer absences and disciplinary issues because students who feel valued and included tend to be more invested in their positive participation in learning. For parents and caregivers, inclusive classrooms mean that their children are spending time in a nurturing space.

### Inclusion in the classroom

Creating an inclusive classroom is not rocket science. Educators need to be sensitive towards all their learners and must ensure that the classroom is a safe space for all the students. Here are some ways in which this can be done:

**Choose texts astutely**

At the beginning of the school year, it is a good idea to ensure that the materials that are going to be used cater to a variety of geographies, cultures, communities, ethnicities, languages, etc., from across the country. If prescribed materials focus on a particular culture, educators must ascertain that their discussions push the envelope and allow for other perspectives to be included. For instance, if in a language classroom, students are reading a story about a family celebrating the Pongal festival, educators can tell students about other harvest festivals and encourage them to share how they celebrate such festivals at home. Integrate these enriching conversations into their everyday lives and avoid the "tourist approach"* when it comes to dealing with diversity.

*The term refers to engaging with a culture in a superficial, external way, often treating it as an exotic or isolated concept rather than understanding it as part of everyday life.

**Use accessibility features**

If digital materials are being used in the classroom, inclusion must be thought about at the time of conceptualization itself. There are many assistive/ adaptative technologies and accessibility features easily and freely available to help learners with special abilities and learning preferences. Educators can make use of the following ways to help learners perceive content as they are comfortable.

- Use screen readers to read aloud web pages for learners who cannot read the text. Educators can integrate free or built-in screen readers like Microsoft's Narrator or ChromeVox. They can start by using these tools to read aloud digital content like web pages, e-books, presentation slides, and documents.
- Use screen magnifiers and customized fonts in different types, colours, spacing, and sizes to help students with low vision.
- Use speaker notes, captions, and audio and video descriptions to help students with hearing impairments. They should enable captions for videos (e.g., YouTube's built-in captions) and include speaker notes in presentation slides. For audio content, they could use transcription services like Otter.ai to provide written content for students with hearing impairments.
- Simplify the digital classroom environment by using tools like Google Classroom's "minimal view" to remove distractions. Create clear, concise presentations that focus on key information only, and avoid overcrowded slides or web pages.
- Provide alternate presentations of content. The text should include translations to other languages including sign language and braille. Offer translated materials or basic sign language resources through free tools like Google Translate, ensuring inclusivity for non-native speakers or those with hearing impairments.
- For students with mobility challenges, encourage the use of voice-to-text software such as Google Docs' voice typing feature, or speech-to-text tools in smartphones. Students can also use built-in text-to-speech features on phones or computers for easier interaction with materials.
- Allow tactile, auditory, and visual notifications by prompting the learners with sound, vibration, visual dialog boxes, etc. Set up visual or auditory cues in presentations (such as using sound alerts for quizzes) or digital classroom management tools (like reminders for upcoming tasks) to ensure students stay on track and are notified in different ways.
- Educators can use free online tools like WAVE (Web Accessibility Evaluation Tool) to check their class websites, digital documents, or presentations for accessibility. This helps identify any issues that might prevent students from fully engaging with the content.

**Circumvent pigeonholing**

Stereotypes based on culture, religion, or gender must be avoided at all costs. Nowadays, most publishers are conscious of this while creating their textbooks and other learning materials. Educators must also take this seriously and be more inclusive in their explanations and examples. For instance, mothers should be shown to have independent careers, sit-at-home fathers taking care of the home and kids should be normalized, and new families with same-sex couples, co-parenting, reconstituted families, pet parents, plant parents, and single-parent families must be showcased. Educators must allow for healthy discussions on these topics and contextualize them as far as possible.

**Select neutral pronouns**

Pronouns are linguistic tools that we use to refer to people. Consequently, it is critical that the people being referred to identify with those pronouns and are allowed to point to the pronoun that they want to be addressed by. Using incorrect pronouns may lead to the person feeling alienated, dismissed, and disrespected.

Also, assuming someone's pronoun just by looking at them is a harmful practice and implicitly reinforces stereotypes. It is best to use gender-neutral pronouns. For instance, 'they', which we have used as a plural pronoun is being accepted worldwide as a singular pronoun.

**Model all-encompassing language**

Inclusive communication is key to affirming diversity. All instructions whether printed or verbal must eliminate negative biases. Remarks such as 'Don't cry like a girl', 'Act like a lady', or 'Man up' must be avoided. Instead, inclusive terminology must be used consciously. Here are some commonly used presumptuous words along with their alternatives:

- Mothers and Fathers → Families, caregivers
- Boys and Girls → Students, scholars
- Businessman → Businessperson, business representative
- Chairman/chairwoman → Chairperson, chair
- Salesman → Salesperson
- Manpower → Workforce, workers
- Postman → Letter carrier, postal worker
- Manned → Crewed
- Man-made → Human-made
- Mankind → Humankind, humanity
- Air Hostess → Flight attendant
- Waiter/waitress → Server
- The common man → The average person
- Husband /Wife → Spouse/partner

Also, groups based on gender must be replaced with those based on the first letter of students' names, their birth number, etc. Euphemisms or agreeable expressions substituted for potentially offensive words should be used. For example, use 'economically disadvantaged' for 'poor', 'pre-loved' for 'second hand', 'visually impaired' for 'blind', 'passed away' for 'died' and so on.

**Differentiate and scaffold instructions**

Educators must adjust their teaching to meet the needs of all students by considering four key factors: content (what students will learn), process (how students will engage with the content), culminating projects (tasks that help students apply and extend their learning), and the classroom environment (how it works and feels). To support all learners, activities and instruction should be modified using tiered approaches, meaning students work on the same important skills and understandings but at different levels of support, challenge, or complexity. Additionally, giving students different amounts of time to complete tasks provides extra support and helps them succeed.

**Pre-assess pedagogical practices**

Before beginning your teaching for the year, assess each student's learning level. You can do this by creating an environment that is personalized to students' needs and by talking about learning that focuses on what students can do and what they would like to do next. This can be done through tutorials, Individual Learning Plans (ILPs), and short and long-term goal setting by the learner so that they feel they have ownership of their learning. If you provide students with opportunities to tell you what is working and what needs attention, you will have a better idea of what to focus on. Also, giving students a choice is empowering in many ways because it creates equal and fair opportunities to show learning and progress in a way that a standard test (which many students struggle with) may not.

**Create suitable learning environments**
Design learning spaces keeping in mind both inclusivity and functionality. Ramps for students with wheelchairs, cluster seating, and distance from the board should all be things to consider. Determine how you can arrange your room to accommodate kids with sensory difficulties, mitigate distractions, and make learning easier for specially-abled students. Also, make sure that classroom posters, pictures, books, toys, and other materials are diverse in terms of race, ethnicity, gender, age, family situations, and so on. Varied representations are not only important for making diverse student populations feel included but these components also have an integral role in keeping students focused on their studies rather than in navigating the classroom.

**Stand up for social justice**
Do not ignore prejudiced behaviour in your classroom. Avoiding the problem will not make it go away, and your silence may even give the appearance of tacit approval. Instead, set clear expectations of socially-inclusive behaviour and make it clear that you will not tolerate racial, ethnic, religious, sexual, or other offensive jokes, or slurs, and explain why. If you cannot respond at the time an incident takes place, respond as soon after as possible before the problem worsens. For the students who have been humiliated or "othered", the best approach is to be compassionate so that they see themselves as valued participants in the classroom. Make empathy training as central to your lesson plans as reading, writing, and solving sums. You should be a role model for students when it comes to standing up against discrimination. Reflect and practice inclusive multicultural values in all aspects of your life, not just inside the classroom. Demonstrate that you respect and value everybody's knowledge, talents, and diversity.

**Handle harmless disruption delicately**
While it is important to enforce and reiterate certain standards of behaviour in the classroom, educators must deal with small bouts of disruption sensitively. For instance, instead of writing the name of a child who is unable to follow a repeated instruction on the board, you can counsel them later in a personal interaction. Calling out one student is singling them out and can be humiliating, especially if they are dealing with attention deficit disorders. Also, it is a good idea to keep parents and caregivers informed and involved at all times so that they are aware of what their children are experiencing in their school life.

In conclusion, according to UNICEF, disabilities are one of the most serious barriers to education around the world. Inclusive education aims to break down this barrier by recognizing and promoting all children's right to education, and prioritizing their wellbeing, autonomy, and dignity. Thus, it becomes pertinent for school authorities and educators to strategize ways of developing equitable and equality-focussed learning environments through structured conversations, awareness of social and emotional conditions, alternate means of participation, specific classroom norms, personalized interactions, non-competitive ways of learning, cross-cultural communication, inclusive language, and sensitive conflict resolution.

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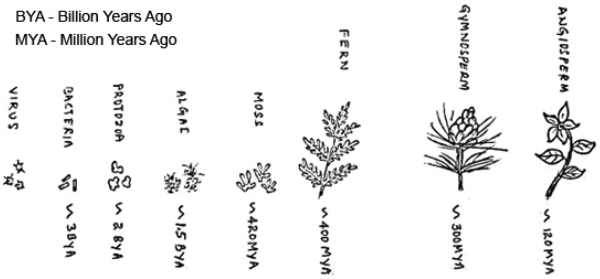
The author is an English language curriculum designer and editor. She holds a Master's degree in English from Lady Shri Ram College for Women, University of Delhi, and a CELTA from the University of Cambridge. Her extensive experience includes serving as an English language assessment specialist for Trinity College London, assessing various international English language examinations, leading examiner and marker cohorts for Qualifications and Assessments International, UK and developing educational materials for the K-8 segment. She has authored two series – *Grammar Sparkle* (Grades 1-8) and *Let's Learn to Listen and Speak* (Grade 4) and is currently writing her next. As a consultant editor with various renowned publishing houses, she has edited over 100 books ranging from academic to fiction, non-fiction, poetry, and children's writing. Her articles on ELT pedagogy and learning strategies have been published in several educational magazines and blogs. She can be reached at <sanjheegianchandani28@gmail.com>.

# Looking into a microscopic world

Nandini Dholepat

Sitting in the Gymnosperm garden, Anand and Vivan discussed with each other and drew a diagram of simple to complex life forms. Raja added a timeline to the diagram.




Photos: Nandini Dholepat

"Great! This gives clarity on how different forms of life evolved on Earth. So, with this evidence, we now have come to an understanding that human beings are highly evolved, we are not struggling for survival and are fighting for our greed and not need," said Madhavi.
"True, if we neglect these less understood forms, which support the survival of all other higher organisms, we will lose everything soon," I replied.
"But how? We are growing ferns in gardens," Rini commented innocently.

Raja tried to explain, "In natural ecosystems, algae, bryophytes, and pteridophytes play a major role. They provide food for small animals. Their continued presence in the ecosystem confirms the existence of pure water, fish, mussels, prawns, and other such dependents. They are all very delicate. If the water gets contaminated or the water flow reduces, plants will wither away and animals and insects will eventually vanish."

"There was a time I used to think that all this greenery was only ornamental and not of much use beyond that. Our approach towards human-centered conservation should change," Sujatha said.
"Let's get a move on. I got a message from my colleague Jayanthi, we can go to the lab," Raja said.
"What is this? Looks like a tiny fruit," Vivan pointed at a plant at the entrance of the staircase.

This is 'Taxus', it is a gymnosperm," I replied.

"It was brought from Arunachal Pradesh, from a high altitude. Fortunately, it is surviving even in this environment. It is actually a tree, but as it is pruned regularly, it doesn't grow tall. And what you think is a fruit, is a scale. It is a protective covering over

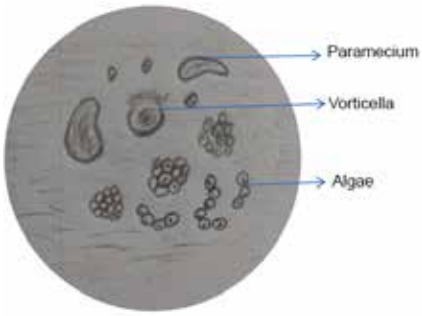
the bud, that turned out fleshy here. You can see an opening and a seed inside. Birds eat the seeds and help in their dispersal," said Raja.

Inside the building, we were asked to sit on a bench. Frames of famous scientists adorned the wall. Anand pointed to a photograph, "See, this is Alexander Fleming, he invented Penicillin. The same picture is in my textbook."

"I know the person in the left photograph. It is Charles Darwin, who proposed the theory of evolution," Rini also showed off her knowledge.

"Hi everyone, I am Jayanthi. I am a researcher in the microbiology lab. Before we go in, please keep your bags in these cupboards. Just carry your notebooks and pencils, and remove your footwear and wear these." Jayanthi pointed to the slippers kept for visitors. Then, she asked us to wash our hands and gave us face masks. She led us to the lab and informed us about their regular activities. She also showed us how to place a specimen on the microscope and adjust the microscope to observe the specimen.

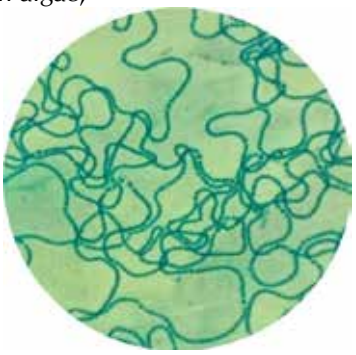
Raja called everyone, "Come here, take a look at this. I have placed a water drop collected from the pond on a glass slide. Try and sketch what you see in your books."




Jayanthi asked me to help the children look through the microscope so that she could get the next set of slides ready.

Shalini said, "I saw some green round cells, something like a bean, which was moving, and something like a round cup with hair." Everyone agreed with her.

I marked the picture to explain. "Paramecium is a single-celled organism like Amoeba; it is bean or slipper-shaped. The cup-shaped structure is Vorticella. Both are protozoans, meaning first-formed animals. They are Eukaryotes, which means they have proper nuclei in the cell. Both belong to the ciliates group, which means they have hair-like structures on the body, which helps in movement. Vorticella are called 'epibionts' as they live on the bodies of other animals, like mosquitoes, prawns, and tadpoles. They are not harmful to the host. It is a neutral relationship. And the chain-like and clumped forms you see are green algae,"



"I focused only on the algae on this next slide. It looks like a chain of beads," Jayanthi said.

"Now let us move to the next microscope, one by one, please. Can you see these structures when you look through the microscope?" I showed a pencil sketch to the children.
"What are these? It looks like a whip," Anand said.
"It looks like a horn on a brick," Shalini said.
"These are hairs or trichomes on the fern leaf," I replied.
"Oh! This causes itching. I remember now," Anand replied as he looked through the microscope.

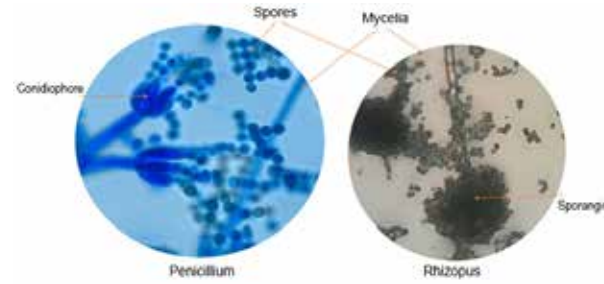
Jayanthi called us to the next table. "See, this is agar powder, we use it to prepare food for growing bacteria and fungi. From Agar, culture media is prepared and poured on a petri plate to help in the growth of bacterial and fungal colonies. This is an air-exposed plate. Here, you can see the growth of different bacterial and fungal colonies."

"We know Agar Agar, it is a seaweed," Rini said.
"Very good. It is from the same source, but this is laboratory grade," Jayanthi patted Rini.
"Look at these specimens!" Rini pointed to a lemon and some sprouts that were rotting.

"Now, let us observe fungi. The green mold on the lemon is Penicillium. It is commonly found on citrus fruits. The black cottony growth on the sprouts is Rhizopus. The fungi life cycle involves spore production, either sexually or asexually. The spores then germinate and grow into mature fungi, which eventually form mycelium. This mycelium then forms the fruiting bodies that produce and disperse spores to start the new cycle," Jayanthi explained.

"The fungi life cycle is similar to moss and fern life cycle, is it not? Earlier Nandini had explained that to us," Sujatha said.
"We add certain specific stains so that the different fungi and bacteria can be easily identified," Jayanthi said.

Jayanthi adjusted the specimen under the microscope and drew pictures on a white sheet. She marked the pictures to explain.




"These small round forms are conidiophores and the black round forms are sporangiospores. If all are spores, why the different terminology?" Anand asked.
"Conidia are borne singly or in chains on special branches called conidiophores. The sporangiospores, on the other hand, are contained within the sporangium. They are generally produced in large numbers. The slender tape-like structure below is mycelia," Jayanthi replied.
Rini said, "We have seen fungi, but where are the bacteria? Can we see bacteria under the microscope?"
"Sure, wait for some time. I will get the specimen ready," Jayanthi replied and went away to prepare the next few slides while we busied ourselves discussing what we had seen so far.

**Suggested activities**

1. What are the different types of relationships you observe among organisms?
2. Identify the bacterial and fungal infestations you observe in your kitchen.

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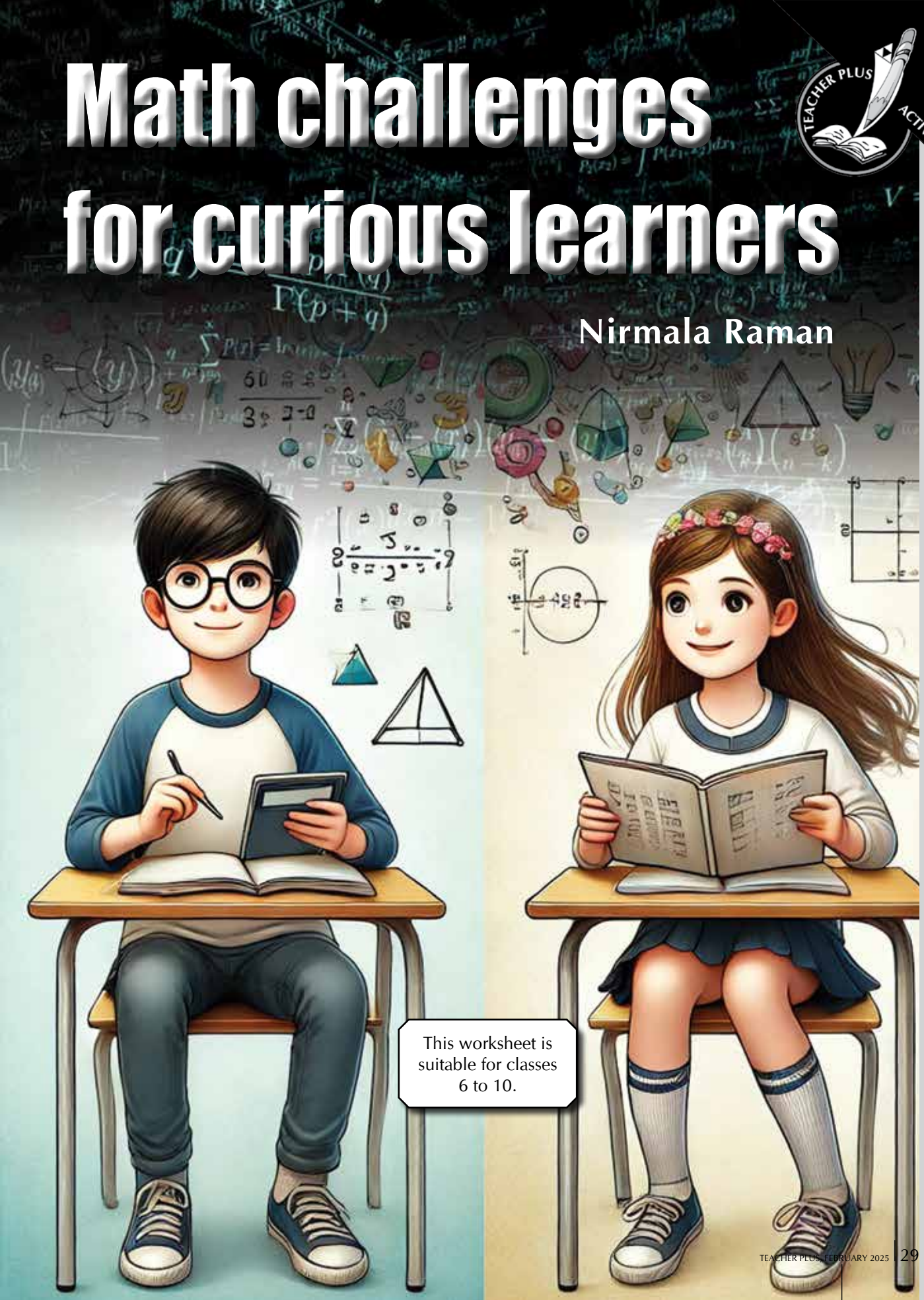
# Math challenges for curious learners

Nirmala Raman



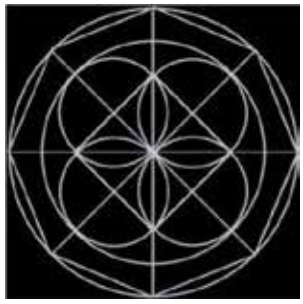


This worksheet is suitable for classes 6 to 10.

# Activity 1: Angles and Symmetry

**The geometry of defense**
A ***dojo*** is a training place for martial arts. The word ***dojo*** means **"the place of enlightenment"** in Japanese. It's a place where people grow mentally and physically through martial arts training, aiming for personal development.

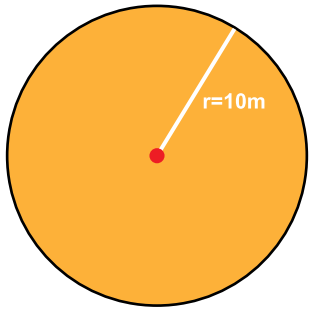
Martial arts training in a *dojo* involves several mathematical concepts, especially in geometry and angles. These concepts help improve the learners' techniques and movements.



Movements like strikes and defenses involve precise angles for effectiveness, such as 45°, 90°, or 180°. Opposite or complementary angles often play a role in countering attacks.

The dojo often has a symmetrical layout, helping to teach balance and spatial awareness. Techniques involving spins use degrees, similar to the concepts learned in rotational geometry. This aligns well with how mathematics applies to physical training.

Ruthikaa, a martial arts prodigy, had always dreamed of mastering the art of strategy. Her coach, Sensei Achintyasan, welcomed her into an elite training session. The dojo, a perfect circle with a 10 meter radius, became her testing ground.




"Ruthikaa," Sensei began, "strength alone is not enough. To defend effectively, you must understand the geometry of combat. Every move in battle is an angle, and every angle can be countered with precision."

**Questions**
- If an attack comes from 135°, where should Ruthikaa position her defense?
- For attacks at 90°, 225°, and 315°, what are the corresponding defense positions?
- If an attack occurs at 60°, which defense position (0°, 90°, 180°, or 270°) is closest?

**Challenge**
Sensei adds, "If moving 1° takes 2 milliseconds, how long will it take to reposition from 45° to 270°?"

Ruthikaa realized geometry isn't just about shapes – it's about strategy.

## Activity 2: Triangular numbers

**The art of flower garlands**
The streets of a South Indian village were alive with preparations for the annual temple festival. Veda and her grandmother, Amma, sat at a flower stall, designing garlands for the deity.

"To make a beautiful garland, we must balance jasmine and marigold flowers while keeping it symmetrical," Amma explained. Ravi, Veda's cousin, eagerly joined to help with the math.

**Math problem**

- **Length:** The garland is 2 meters (200 cm) long. Each flower is 7 cm with 5 cm spacing. How many flowers are needed?
- **Types of flowers:** If the pattern alternates between jasmine and marigold, how many of each will be used?
- **Circular arrangement:** If the garland is made into a circle, how will the flowers be split on each side?

## Activity 3: Probability

**The handshake event**
Akshara School's annual Handshake Challenge was in full swing. Excited chatter filled the air as students lined up to greet each other. Sesha, known for his quick calculations, observed the activity closely.

"Did you notice that some students keep their thumbs up during handshakes?" Sesha asked his friend Archana. "I wonder how many of them do that. If 138 students are here and about 60% keep their thumbs up, can we figure out the number?"

**Questions**

1. How many students likely kept their thumbs up?
2. What's the probability that exactly 83 students did so?

## Activity 4: Repeated addition and pattern

**The clock maker's mystery**
The air hung thick with the scent of dust and old paper as Gayathri and Vishal delved deeper into their grandfather's attic, a treasure trove of forgotten memories. Their eyes widened with wonder as they stumbled upon a peculiar timepiece, its ornate frame glinting in the dim light. It wasn't just a clock; it was a riddle wrapped in brass and gears.

"Look at this!" Vishal exclaimed; his voice filled with excitement. "It's a clock that doesn't tick like a normal one. It skips forward by 3, 5, 7, and 9 hours!"

Gayathri's curiosity was ignited. "That's fascinating!" she said, her eyes sparkling. "Let's figure out how it works. If it starts at 2 o'clock, how long will it take to show 2 again?"

As they leaned in to examine the intricate mechanism, a sense of adventure filled the air. They knew that solving this puzzle would unlock a piece of their grandfather's past, and perhaps, a bit of their own future.

Challenge: If another clock jumps 4, 6, 8 hours, etc., when will both clocks align?

## Activity 5: Logical reasoning, trial and error

Often, we use a combination of logical reasoning and trial and error to solve math problems. For instance, when solving word problems, we use logical reasoning to break down the problem into smaller steps. Then, we might use trial and error to test different approaches or values.

By understanding and applying logical reasoning and trial and error, you can become more confident and effective problem-solvers in mathematics.

**The chocolate puzzle**
Ananya's heart pounded with excitement as she stepped into Mr. Jose's enchanting candy shop. The air was thick with the sweet aroma of chocolate and the shelves were lined with an assortment of delectable treats that seemed to stretch to infinity. She knew that behind the shop's whimsical facade lay a world of mathematical puzzles, waiting to be unraveled.

Today, Mr. Jose had a special challenge for her. With a mischievous twinkle in his eye, he presented a shiny tray of 28 chocolates. "Ananya, my dear," he said, "I want you to divide these chocolates into 5 groups, but each group must have a different number of chocolates. Can you crack this sweet little puzzle?"

Ananya's mind raced as she pondered the problem. She knew that this wasn't just about dividing candy; it was a test of her mathematical skills. With a determined look, she grabbed a pencil and paper and began to work. She knew that the key to solving this puzzle lay in finding the right combination of numbers that would add up to 28.

You are Ananya now!

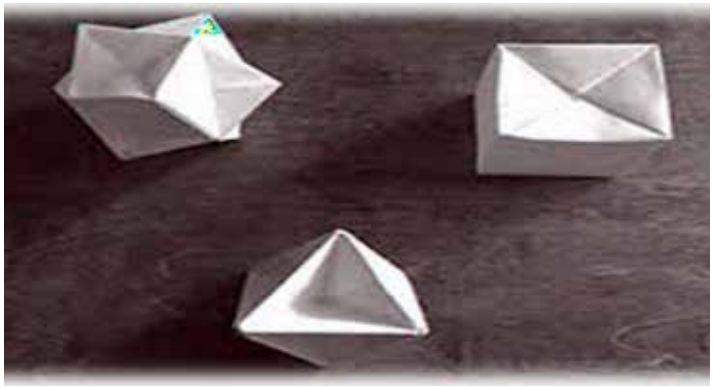
Ananya's thought process (to guide the student subtly):

"If each group needs to be different, I could try starting small. May be one group has 1 chocolate, the next group has 2, and so on. But will it add up to 28? Let me test it."

*Logical thinking sweetens challenges!*

## Activity 6: Geometry, fractions and proportions, measurement

**Origami math**
Origami is a perfect way to see geometry in action! When you fold paper, you're creating shapes like triangles and squares and using symmetry to balance your design. Each fold involves angles, so it's like you're practicing math without even realizing it. Origami also transforms flat, 2D paper into 3D objects, like animals or flowers, which is fascinating. The crease lines left behind are like a map of your folds, and studying them helps you understand how design works. Plus, origami inspires real-world ideas, like folding solar panels or medical devices!



**The Origami championship**
The excitement was palpable as Ravi and Maya sat down, their minds racing with the challenge ahead. The Origami Championship was just around the corner, and this year's contest promised to be the most challenging yet. The organizers had thrown a curveball, a puzzle that required a blend of artistic skill and mathematical precision.

With a determined glint in their eyes, Ravi and Maya unfolded their 24 cm square sheet of paper. They knew that each fold would dramatically reduce the area, but would it shrink to a point where crafting a graceful crane would become impossible? The answer lay in the intricate dance of numbers and geometry.

**The challenge was unique**
1. How many folds can they make before the paper becomes too small to fold?
2. What are the dimensions and area of the paper after each fold?
3. After 4 folds, will they have enough paper left to make a crane that needs at least 5 $cm^2$?
4. Can Ravi and Maya fold the paper into smaller sections to make multiple cranes? How many cranes can they make and how much paper will be leftover?

"Let's figure this out," Ravi said, carefully measuring the paper after each fold.

**Questions for you to answer**
1. How many folds can Ravi and Maya make before the paper becomes too small to fold?
2. What are the dimensions and area of the paper after 4 folds?
3. Can they make one crane after 4 folds?
4. If they decide to make multiple cranes, how many cranes can they make? How much paper is left?

*Folding paper makes math fun.*

**Bonus challenge**
What if Ravi folds the paper diagonally? How does this affect the area and number of cranes they can make?

## Activity 7: Percentages

**Learning about discounts and percentages**
**Discount math**
Your parents are planning to buy you a smart phone for your birthday. While browsing, they come across a great deal.

"The phone costs ₹30,000," your father says, "but it has three successive discounts of 25%, 10%, and 5%. Let's see if you can figure out the final price and how much we save overall."

The air crackles with anticipation as you delve into the world of numbers. You know this is your chance to shine, to prove that math isn't just a subject in school, but a powerful tool that can unravel even the trickiest of real-world problems.

With a determined glint in your eyes, you begin your calculations to work out the solution.

## Activity 8: Ratio and Proportion

**A ratio is a comparison of two quantities.**

**A proportion is an equation that states that two ratios are equal.**
Imagine you're making a delicious milkshake. You have a recipe that calls for 2 scoops of ice cream and 1 cup of milk. This is a ratio of 2:1. This means for every 2 scoops of ice cream, we use 1 cup of milk. If you want to make more milkshake, you can double both the ice cream and milk, keeping to the same ratio. So, 4 scoops of ice cream and 2 cups of milk is also in the ratio of 2:1.

Recipes often use ratios. If you double a recipe, you double all the ingredients, maintaining the same taste.

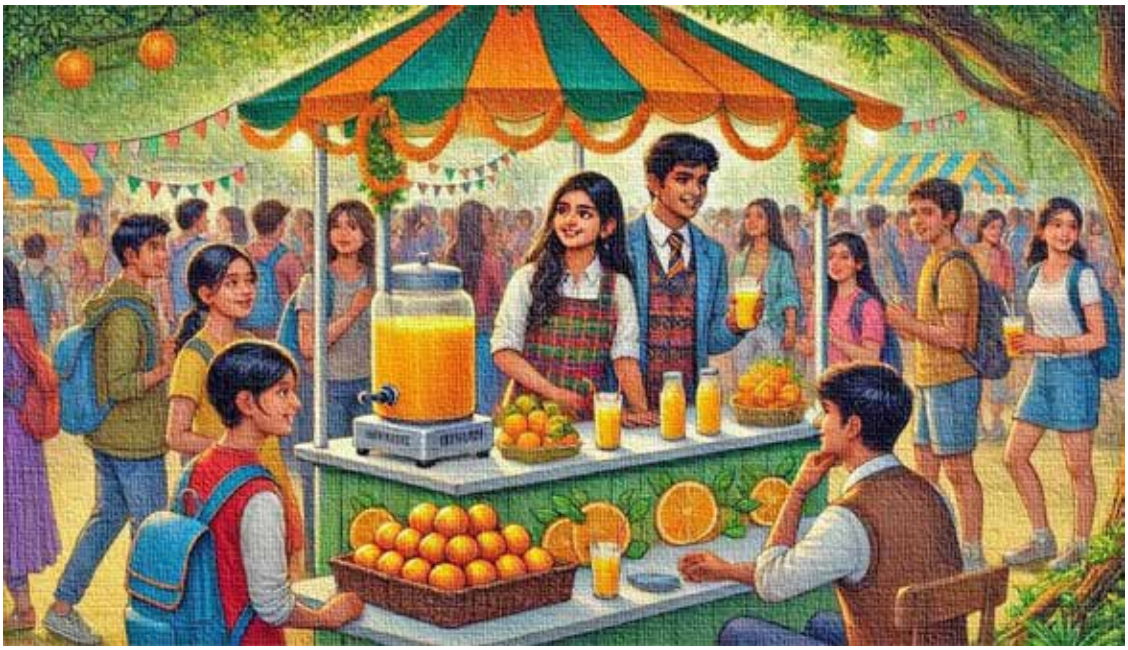
Maps use scales, which are ratios. For example, 1 cm on a map might represent 10 km on the ground.

To create a specific shade of paint, you need to mix different colours in a certain ratio.

Businesses use ratios to analyze their financial health, such as the debt-to-equity ratio or the profit margin.

By understanding ratios and proportions, we can solve many real-world problems and make informed decisions. So, let's explore the fascinating world of ratios and proportions!

**Juice mix-up**
Ravi and Priya run two juice stalls at a local festival, and they both prepare orange juice by mixing orange concentrate and water. However, they use different recipes:



- **Ravi's recipe:** 3 liters of orange concentrate with 7 liters of water.
- **Priya's recipe:** 5 liters of orange concentrate with 5 liters of water.

A customer wants to know:
1. Which juice is more concentrated with orange flavour?
2. If both Ravi and Priya prepare **20 liters of juice** each, how many liters of orange concentrate does each use?
3. Who uses **less water** for the same amount of juice?

Help the customer figure out which juice has a stronger orange flavour and which stall uses less water for 20 liters of juice.

**Hints for students**
Find the proportion of orange concentrate to total juice for both recipes:
Concentrate Proportion = Concentrate / (Concentrate + Water)
1. Multiply the proportions to adjust the recipes for **20 liters** of juice.
2. Compare the amount of water used by each stall.

## Activity 9: Fractions

Whether you're a chef, baker, or home cook, chances are you've had to incorporate fractions in real life and in the kitchen beyond the use of measuring cups and spoons. Fractions represent a portion of a whole. For example, one quarter is equal to 25% of a whole, and a half of that is one-eighth of that whole. Fractions may come particularly handy when cooking as you come across recipes that don't represent the number of servings you need to prepare or other instances where you may have to do the math on the fly. For example, if you're

cooking for four people and the recipe was developed for eight people, you'll need to correctly divide all your ingredients in half or you'll end up with too much food – or a dish that didn't come out right because you added too much of one ingredient.

**Baking fractions**
Riya's mouth watered as she imagined the warm, buttery aroma filling the kitchen. Her grandmother's chocolate chip cookies were legendary.

But today, they had a special mission: to bake a dozen cookies, just the right amount for their upcoming picnic, but the recipe they used was for 16 cookies.

"How can we shrink the recipe without ruining the taste?" Riya pondered, her eyes scanning the card.

Her grandmother, ever the wise baker, smiled knowingly. "A little math can make our baking adventure even sweeter," she teased her, “Let’s work together to figure it out."

**The original recipe includes**
- 2 cups of flour
- 1 cup of sugar
- 3/4 cup of butter
- ½ cup of chocolate chips

Riya needs to adjust the recipe to make exactly 12 cookies instead of 16.

**Bonus challenge**
If Riya decides to make only 8 cookies, how would the ingredient measurements change?

## Activity 10: Discreet and continuous

The All-Star Challenge is in full swing, and you, as the coach, need to strategize how Steph Curry, Kevin Durant, and LeBron James can score exactly 100 points for the team in a 48-minute game. Here’s the catch:

- Steph Curry is amazing at 3-point shots.
- Kevin Durant excels at 2-point shots.
- LeBron James is known for making free throws, each worth 1 point.

Your task is to find how many shots of each type each player should make to ensure the team scores exactly 100 points.

How many shots of 3-pointers, 2-pointers, and free throws will the team need to take to get to exactly 100 points? You can break it down into a combination of shots.

Now imagine you have only 48 minutes for the game. Each shot takes exactly 1 minute. How can you allocate the 48 minutes efficiently, so the team scores exactly 100 points?

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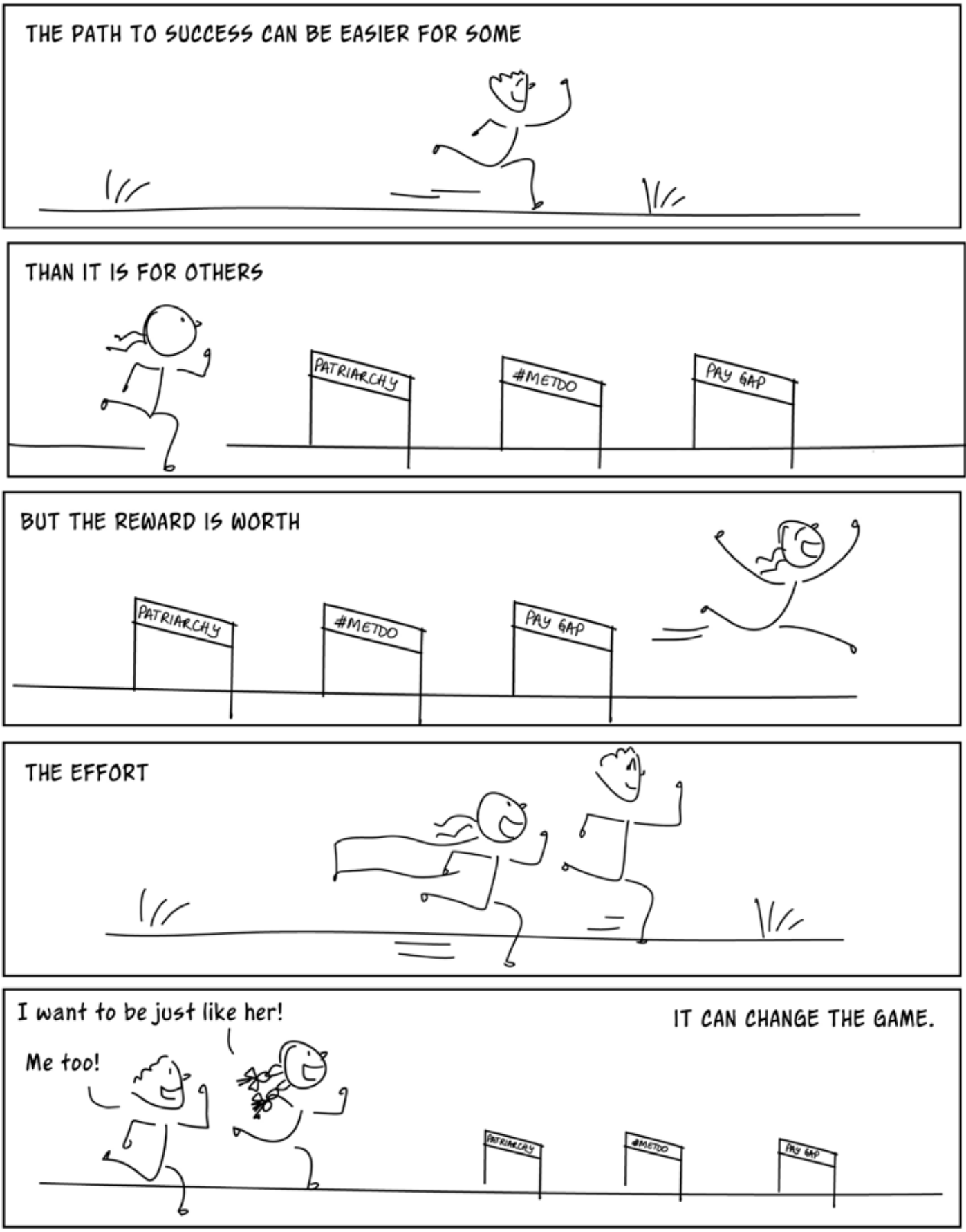
# THE LIFELONG LEARNER: BRIDGING THE GAP

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# Transforming math education through a global lens

Monica Kochar

Image generated using AI

In today's world, where digital technologies and global dynamics reshape human interactions, the need for a global mindset has never been more critical. This interconnectedness compels educators to rethink traditional approaches and prepare students for an increasingly complex world.

## Moving beyond traditional models

The industrial-era model of education, designed for uniformity and standardization, no longer meets the needs of today's diverse classrooms. Students, today, come from varied cultural, linguistic, and socio-economic backgrounds, bringing unique perspectives and challenges to the learning environment. Yet, many educational systems remain tethered to outdated methods that fail to accommodate this diversity.

Dighe (2000) underscores the urgency of moving away from the "one-size-fits-all" approach, advocating instead for flexible educational frameworks that address the unique needs of learners. To meet this challenge, we must embrace global competencies – knowledge, skills, and perspectives that enable students to understand and engage with global issues.

This shift requires not only curriculum changes but also a transformation in how we view education itself.

## Integrating a global mindset in mathematics

Mathematics, often regarded as an abstract and isolated discipline, holds immense potential for fostering a global mindset. When contextualized

with actual global issues, mathematics can become a powerful tool for nurturing critical thinking, empathy, and problem-solving skills.

Incorporating global themes into mathematics lessons allows students to connect abstract concepts with real-world applications, fostering both academic and emotional growth. Below, we explore examples of mathematics lessons that integrate global issues to develop a global mindset.

### Geometry: Cultural patterns across the globe

Geometry provides a window into the artistic and cultural heritage of societies worldwide. For example, Islamic mosaics, African textiles, and Native American beadwork all showcase intricate geometric designs. A lesson might begin with the question: "Why is geometry so prevalent in art, and what cultural meanings does it hold?"

**Activity:**

- Students research geometric art from different cultures and create their own designs inspired by these traditions.
- They analyze the geometric properties of the designs, such as symmetry, tessellation, and transformations.
- As a culminating project, students apply these principles to create art installations for their school or community.

This lesson connects mathematical concepts with cultural appreciation, encouraging students to see mathematics as a universal language.

### Number systems: Tracing cultural contributions

A study of historical number systems offers insights into how civilizations have shaped mathematics. From the Egyptian and Babylonian systems to Mayan numerals, students can explore the evolution of numerical representation.

**Activity:**

- Students compare ancient number systems with modern ones, analyzing their strengths and limitations.
- They investigate how these systems influenced trade, astronomy, and culture.
- As a creative project, students design a fictional civilization with its own number system and present it to the class.

This activity highlights the interconnectedness of cultures and the role of mathematics in human progress.

### Data Analysis: Understanding population dynamics

Data handling offers a platform for exploring global issues such as population growth, urbanization, and sustainability.

**Activity:**

- Students analyze population data from different countries, creating graphs and charts to visualize trends.
- They discuss the factors driving these trends, such as migration, birth rates, and economic development.
- In groups, students propose solutions for managing population challenges and present their ideas to the class.

This lesson bridges mathematics with critical global issues, encouraging students to think creatively and collaboratively about possible solutions.

### Percentages: Food security and nutrition

Percentages can be used to examine disparities in food access and nutrition between urban and rural communities.

**Activity:**

- Students collect data on food consumption patterns and calculate percentages to identify trends.
- They explore how factors like income, education, and geography influence dietary habits.
- As a final project, students create awareness campaigns promoting healthy eating in their communities.

This lesson connects mathematics to personal and societal wellbeing, empowering students to address real-world challenges.

### Holistic assessment for a global mindset

To assess the development of a global mindset, educators must go beyond the traditional metrics of mathematical accuracy. Holistic assessments should measure students' ability to apply mathematical concepts, engage with global issues, and reflect on their learning. For example –

**Geometry:** Students submit a reflective essay on the cultural significance of their designs alongside their mathematical analysis.

**Population dynamics:** Students present their findings and solutions to peers, receiving feedback on both their mathematical and critical thinking skills.

In this way, by incorporating diverse assessment methods, educators can ensure that students develop both academic proficiency and global competencies.

## Conclusion

Incorporating a global mindset into mathematics education transforms the way students perceive and engage with the world. By connecting abstract concepts to real-world issues, we can inspire curiosity, empathy, and a lifelong love of learning.

Reflecting on my own experiences, I recall a class where students initially dismissed mathematics as irrelevant. When I introduced lessons linking math to global issues, their attitudes shifted dramatically. Suddenly, they saw the subject as a tool for understanding and solving real-world problems.

As educators, we have the power to ignite this spark in every student. By weaving global themes into our lessons and assessments, we can prepare them not only for academic success but also for meaningful contributions to a connected world.

The author is a seasoned educator and curriculum designer specializing in math education. With over 10 years of experience, she created HumaneMath, a human-centric math program, and has worked with leading organizations in edtech, publishing, and training. Her expertise spans curriculum design, teacher training, assessments, and quality review. Monica has contributed to key startups, including WhiteHat Jr., and mentored several initiatives. She holds a Master's in Education from the University of the People. She can be reached at <humanemath@gmail.com>.

# Can you find

When I was asked to read and write about Jane Goodall's *The Book of Hope*, I couldn't help but think that the request came at the right time. It was the end of last year, which is a great time for some reflection and to look to the new year with hope. It has been difficult to feel hopeful given Israel's ongoing genocide in Palestine and its assault on other countries, politics leaning to the right, continuous threat of communal tensions, and increasing divides between the wealthy and the poor. Goodall and Douglas Abrams, who is the co-author of the book (along with Gail Hudson), touch on almost all these topics in a book that one might assume is about climate activism. This is not to say that the book is not about saving the Earth, it is. But using Goodall's decades of experience fighting a battle that few understand and care about, the authors try to inspire hope in their readers. Goodall and Abrams show us how to be hopeful, especially when the signs of hopelessness are rife.

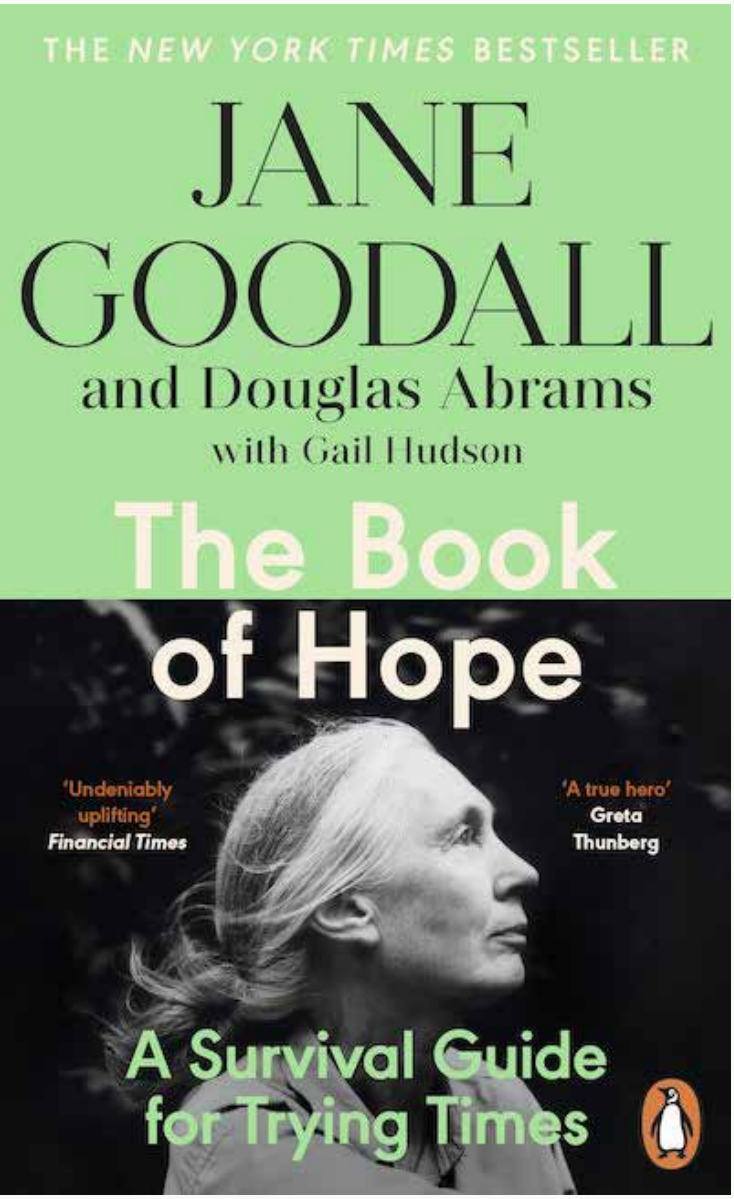
**About the book**

The book, published in 2021, is in the form of a conversation between Jane Goodall and Douglas Abrams. Goodall one of the greatest living naturalists, primatologists, and anthropologists has worked for around 70 years, to raise awareness on the sentience of animals and the need to conserve our natural environment. Douglas Abrams is an award-winning author of bestsellers, with his last being one with Archbishop Desmond Tutu and the Dalai Lama, where they talk about joy. In a sense then, after covering the topic of joy, Abrams is focusing on hope, which is considered to be uniquely human – the ability to desire for an outcome and a belief that the outcome is possible.

Abrams is a skeptic who, through his questions to Goodall, seeks to find what hope is, and how it can be cultivated in children and young people as well as adults. The book begins with Goodall's definition of hope, which is "[that which] enables us to keep going in the face of adversity. It is what we desire to happen, but we must be prepared to work hard to make it so" (pg. 8). There is conversation around agency that allows one to act towards what one hopes for. Goodall provides examples of chimps throwing tantrums for what they want as evidence that hope is not

# hope in a book?

Fiona Vaz




uniquely human. The dialogue between Abrams and Goodall leads her to conclude that hope is a survival trait without which we might perish (pg. 11).

The book proceeds to outline the four reasons for hope according to Goodall. These are (i) The Amazing Human Intellect (ii) The Resilience of Nature (iii) The Power of Young People, and (iv) The Indomitable Spirit. These four reasons for hope are derived from Goodall's experiences of working with chimpanzees in Gombe, Nigeria, and through her years of advocacy and educating people all over the world on saving the environment. The last section provides more insights into the life of Jane Goodall as a messenger of hope. In the next few paragraphs, I take a look at the book, through the lens of a justice-minded reader and assess some of these ideas to see if they have currency today.

**Forms of storytelling**

Goodall firmly believes that statistics and numbers can take one only so far. What makes a real difference, according to her, is the sharing of stories. Hence the book is filled with stories that illustrate not just the four reasons for hope, but also the smaller arguments within each of the sections and sub-sections. The stories are clearly narrated and do the job well of persuading the reader in a gentle way regarding the myriad ways in which hope is demonstrated, built, and shared. The chapters include pictures from diverse sources that also bring the stories to life. Long after one has completed the book, the stories and lessons remain. However, I couldn't help but think that these stories from far flung areas are separated from each other both in time and space. They don't snowball into bigger movements. For example, the Fridays for Future Movement that began with one Greta Thunberg sitting a year in front of the Swedish Parliament galvanized people all over the world to take action for the environment. These protests and resulting activities rattled even strong leaders like Donald Trump and Narendra Modi. One can see Jane

Goodall's work itself as a movement which started conservation efforts in different places, but it feels like these stories do not come together to create hope at the same magnitude as the despair of environmental destruction. Unlike Goodall, I think that statistical data when presented well can move people. For example, the work of Mona Chalabi, a Pulitzer-prize winning writer and illustrator, helps people visualize large scale data and in turn informs public opinion.

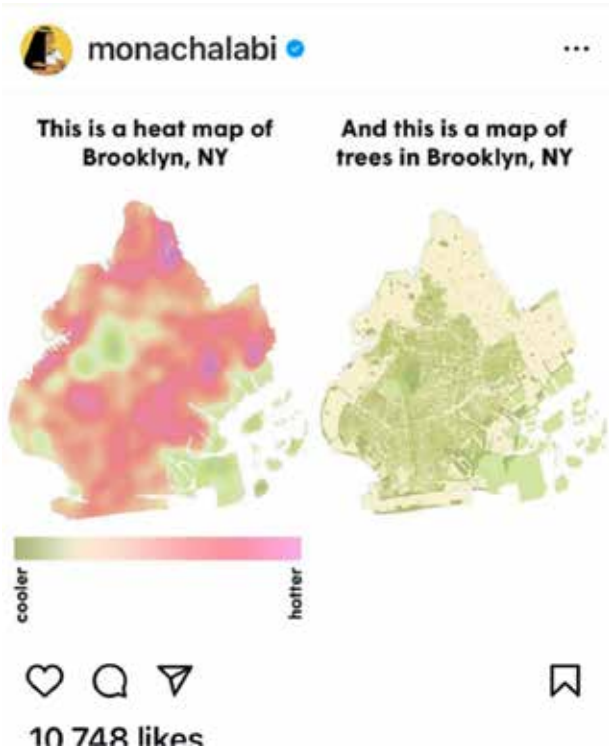



*Figure 1. Mona Chalabi's use of Statistics and Data to convince her audience on the need to conserve nature.*

Chalabi, in her art posted on Instagram, shows through a single image how trees mitigate soaring temperatures in Brooklyn, New York. In another illustration, Chalabi, presents statistical data on the number of natural disasters that have taken place from 1980-2017 and their causes through a compelling visual. It is easy to see that while the incidents of storms have remained the same in this period, extreme temperatures, droughts, and bush fires on account of climate change, have soared from about 200 incidents in 1980 to close to 800 in 2017, i.e., four times the increase in just 40 years. Data can be used to communicate urgency and scale in a memorable way too.

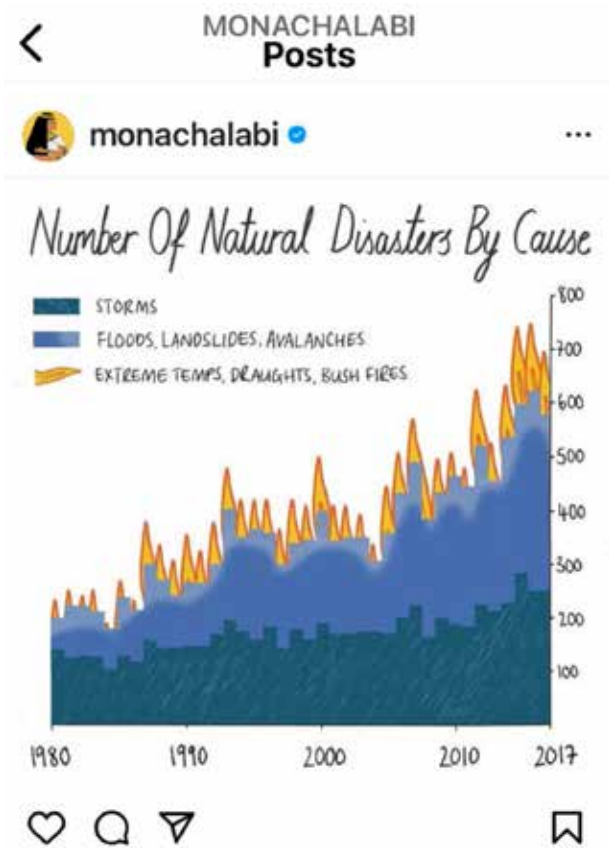



*Figure 2. Mona Chalabi's Instagram Post shows that data can also tell a story.*

Abrams asks several difficult questions throughout the book. He mentions in the beginning that he is a skeptic and uses the conversation with Goodall to build hope. He persists in his questions to Goodall and probes further until he is satisfied with the responses. For example, when Goodall suggests that one of the ways to achieve lasting peace is to develop a new moral code, Abrams is not fully convinced. He writes, "I felt like Jane was leading me step-by-step to a deeper understanding of how we actually might find a better path ahead- though I was also skeptical of any easy solution." (pg. 55) He persists on how one can find a better path to a more compassionate world, when Goodall suggests developing a *new* moral code, which in fact is the old one, *'do unto others as you would have them do unto you'*. Goodall quips that it is easy but proceeds to describe why it has actually been quite difficult to follow. "Greed. Selfishness. Lust for power and wealth." (pg. 55) It appears as if Goodall suggests a fairly complex solution, i.e., of individual people following a moral code, as a simple one. The conversation goes on until there is still more focus on

individual actions and thoughts. Abrams persists, but at some point, allows himself to get convinced even when it is indeed difficult to. The inability to think about others, Goodall states, is because we have lost the ability to think of the future and *others* (p.g. 58).

**Focus on individual efforts**

The narration of stories instead of using data also corresponds to another problem I saw in the book, which is Goodall's focus on *individual* efforts. In Chapter 3, The Resilience of Nature, Goodall and Abrams address eco-grief which is a sense of anxiety, hopelessness, and fatalism among others that might prevent people from taking decisive action. A point that Goodall makes here and in other places in the book is the importance of people playing a role no matter how small. "It is the cumulative effect of millions of small ethical actions that will truly make a difference," (pg. 78) she says. While it is true that *all* people have agency and can catalyze change, Goodall's stories go back to actions such as being aware of what we purchase, going vegan, or examples of change of heart. Maybe it is my own skepticism and hopelessness speaking, but at this juncture of the urgency we face, we need mass actions or policy changes that can make a real difference. Activists all over the world, especially fighting in areas that need public opinion and support focus on key levers that increase public pressure on decision makers such as massive protests and campaigns. Individual changes of heart, such as that of Ben van Beurden, the CEO of Royal Dutch Shell, on account of his 10-year-old daughter's questioning if his company was destroying the planet, made me wonder, how little we expect of people with so much power and influence. At the same time, Goodall's refusal to hold the powerful such as businesses and political actors accountable and focus on systemic change, increased my skepticism. Small individual actions take time, something that most of us, living in extreme precarity, do not have.

**Saviourism and whiteness**

Finally, the conversation between Goodall and Abrams has hints of saviourism, and the privilege of what they would probably never have to experience on account of their social locations – that is of being white (or white passing as I couldn't ascertain Abrams' race). While speaking of cultures different from their own, the language used signals distance. The descriptions of mud floors, and humble homes, and references to places in the United States that are 'worse than developing countries' (pg. 119) show how much Goodall does not really make any of these places in Africa her own. In talking about violence and aggression (pg 50), they refer to choices that reflect which country and culture 'we' were born into that influence our predisposition to violent responses potentially hinting that some countries and cultures are more violent than others. In another line (pg. 159) Goodall mentions, "The same courage and resilience, can be shown by people who have been brainwashed – think of the suicide bombers who believe they will be rewarded in paradise for blowing up innocent people", which can come across as racist, Islamophobic, and xenophobic.

Given my keen interest in gender, I also noticed that Goodall does not mention any woman peer in the entirety of the book apart from Dian Fossey, Birute Galdikas, and Renee Gunther, a teacher in New York. Emmaline Pankhurst, the leader of the Suffragette Movement in Britain and Goodall's mother appear in the book too along with many women who have gained from Tacare, the microfinance work of Jane Goodall Institute. But role models and exemplars, in Goodall's eyes are all male – Winston Churchill, Mahatma Gandhi, Nelson Mandela, her own spouses both of whom were male, and many conservationists, photographers, and thinkers. In the portion on The Spirit That Never Surrenders (pg. 152) Goodall glorifies Churchill's war responses. Although, she speaks of how Britain defended itself from Nazi Germany, militaristic prowess can hardly be an inspiring example in a time of ongoing war, especially when one professes peace towards Nature.

To conclude, I have no doubt that the work of Jane Goodall is commendable and praiseworthy. The thoughts in this book, however, not so much. But this book is about hope – does it mean it left me completely hopeless? I don't think so. Reading the book gave me an opportunity to contrast the ideas of Jane Goodall and Douglas Abrams with the conversation that I am able to witness or partake in today. I feel moved and incredibly hopeful when people today do not mince words and communicate directly with the audience regarding the urgency of the challenges we face. Most of this happens online, and in real life through some citizen-led groups. Unlike what Goodall says that people only need tools to bring about change, I think at this point, the initiatives I see are enabling people with the right knowledge. People and youth, will find the tools. And that makes me hopeful.

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# The role of performing arts in nurturing social-emotional learning (SEL)

Shreshtha Saluja

As a child, I grew up in an education system that placed a greater emphasis on the development of cognitive abilities through math, science, and language. The opportunities to build creative, artistic, and interpersonal learning were limited within the institutional structure. Arts and sports remained on the periphery as extra-curricular or reserved for the 'gifted' students. But arts play a vital role in the holistic development of mind and body. They create opportunities for immersive experiences that facilitate learning with the whole being, instead of engaging only the mind and intellect.

Given the centrality of education in shaping an individual's life, I am curious to find and create methods and pedagogies that are more holistic and experiential. Art, that enhances creativity and sensibilities, can be a potent pedagogical tool to facilitate cross-disciplinary learning, life skills, and social emotional learning (SEL). SEL contributes to the overall wellbeing of the individual and building adaptive skills for a healthy engagement with a dynamic environment. Cultivating social-emotional learning holds potential to engage with systemic issues, ensure mental and emotional wellbeing, and healing through the daily complexities of life. The transformative effect of art integration in education has been corroborated by existing literature (see References) on art-integrated approaches and pedagogy.

Photos: Debosmita Dam

*Children created puppets facilitating cross disciplinary learning of human anatomy and physics concepts like force and energy.*

My position is rooted in affirming the centrality of the body in learning, especially when it comes to social and emotional competencies. It is also pivoted around my own experiences with movement, dance, and theatre. Immersive art engagement has led me to access layered and complex emotions internally and understand the relationship my body holds with external elements – spaces, human, and non-human beings.

## Context

The interconnections between art and SEL have been richly documented in the Western context. In India, the work is still in a nascent stage, but emerging literature around art, inclusion, disability, and social marginalization can be found*.

Farrington and Shewfelt (2020) proposed a theory of action on how arts education can support the social-emotional development of young people. The premise states that art practice inherently carries

*Patricia Benitez Hemans, Rebecca S. Levine, Esmeralda Salas, Amy Bintliff, Caren Holtzman, Carolyn Huie Hofstetter & Gagandeep Kaur (2023) Social and Emotional Learning pedagogy and practices for children living in poverty: teacher perspectives at two Akanksha foundation schools in India, Intercultural Education, 34:6, 533-549, DOI: 10.1080/14675986.2023.2265845.

opportunities for social-emotional learning, such as working with peers in theatre rehearsal, reflecting on one's feelings when painting a mural, etc. The way practice of art can build art competencies, intentionally cultivating the social-emotional components of art education can develop social-emotional competencies.

A similar push for arts-integrated SEL in educational policy is presented by Eddy et al. (2020) through programmes and initiatives** that have successfully demonstrated the use of art as pedagogy for SEL instruction across different regions of the US.

Both these sources stress upon the experiential aspect of arts and how they facilitate integrative learning in embodied and embedded ways.

## Policy in India

The National Education Policy 2020 states development of creative potential, social, ethical, and emotional capacities as aims of education in addition to development of cognitive capacities and advocates for no hard separation (hierarchy) between science and art, curricular and extra-curricular. Adopting a multidisciplinary, holistic and integrated approach to education, it promotes professional development opportunities for educators in latest pedagogies that are rooted in experiential learning and art-integrated approaches.

National Curriculum Framework for School Education (NCFSE) 2023, in line with NEP, places art education under thrust areas acknowledging that it is given inadequate attention and mentions challenges like lack of time, resources, and seriousness given to arts.

**Details of these programmes and initiatives can be found in their paper cited here as Reference 1.

## The performing arts curriculum and the school ecosystem

To build an understanding of how an arts curriculum creates opportunities for experiential learning, gauge the pedagogical approach of the educator, and observe its impact on children with regard to their awareness of self and others, I studied the performance arts curriculum being implemented from grades 1 to 8 in an alternative learning space in Bengaluru. I also attended the year-end performance by children and interacted with them to know and understand their participation and learnings.

The curriculum, based on The Studio Thinking Project framework of the Harvard Graduate School of Education, focuses on building eight studio habits of the mind – essential skills that students must imbibe during the process of performance and creation.

It is important to highlight that the school policy provided a fertile ground for the performance arts curriculum. The focus is on building an ecosystem that enables learning through creation and exploration. The vision and culture emphasize values of sensitivity, compassion, co-existence, and interdependence with the aim to cultivate mutual respect, understanding, and responsibility. The pedagogy is centered around experiential learning, process-centered approach, and cross-disciplinary learning. Art immersion is a prominent part of this school's curriculum.

## The performance showcase

The year-end expo featured performances by children at different stages of learning. Students of grades 2 and 3 showed the transition of a unique creature with superpowers from the egg/fetal position to its full form moving freely across space. They created this performance by crafting the 'creature' from their imagination, giving it character traits, emotions, body language, and movement.

Students from grades 4 and 5 performed a rendition of *The Lorax* by Dr. Seuss, a popular work in children's literature. The story engages with environmental activism and explores the interactions between characters that want to save forests and those who wish to proliferate industries using forest resources. The children engaged with the story to form their own understanding of the social themes explored in it. Based on that, they crafted the performance with dialogue, poem, and characters presenting their narratives. While one group took a cynical and realistic approach, where the conflict isn't resolved and the forest gets destroyed in the end, another group showed resolution, peculiarly with the conflicting parties becoming friends in the end.

## How the performing arts curriculum enabled SEL

The curriculum provided learners with developmental experiences enabling them to act and reflect to make meaning and recognize their own learning. The classroom process involved the use of body and space to express, come together in a circle, listen to instructions and focus, make changes when challenges arose, create the storyline, direct a skit, write creatively, and receive feedback. Imagination, visualization, characterization, engagement with existing literature are all action exercises that allow students to gain emotional awareness of oneself and the world around them. Movement with others in a space, creating a performance together require understanding of rhythm and coordination that enable social-emotional skills like communication, negotiation, and teamwork.

Children with different energies came together for rehearsal and performance. They demonstrated self-regulation, alignment of diverse energies, improved coordination, inclusion, and collective responsibility while working together. They showed increased awareness of self, their thoughts, and ideas. They could articulate their changing moods and emotions and how they responded to these in the classroom. Some children who remained at the sides in the beginning started moving to the center over the course of the programme. Changes in behaviour, such as increased expressiveness, dance, and movement were reported by some parents.

To understand the process behind these changes, I interacted with some students (grade 6) and their teacher and learnt about the mechanisms practiced for conflict management and the role of the educator in the learning space.

## Engagement with conflict and dissonance

The response to conflicts and disagreements can be mapped to action and reflection experiences that have been put in place with intentionality to navigate the diversity of thoughts and ideas. For example, setting up a system where every participant must speak during group processes, learning how to exchange feedback, guiding behaviour through reflective prompts, etc.

The children navigated disagreements by looking for common ground to agree upon, presenting more information to convey their ideas clearly, trying to integrate everyone's ideas to the extent possible, talking and discussing to reach consensus. Some preferred to 'go with the flow' aligning with the energy of the group. In cases where differences could not be resolved, they approached the educator who guided the conversation, instead of giving solutions.

The educator created space for engaging with conflicts and facilitated the process, thereby modelling social-emotional competencies of conversation and reflection. These responses involved stepping out and speaking separately with the conflicting parties, listening to each side, facilitating reflection, empathy, and articulation during dialogue, negotiation through conversation, and facilitating collective reflection on conflict

*Students engaging in exercises in the performing arts class.*

resolution by inviting the group to share inputs on possible solutions. Sometimes, pausing to examine the situation may also be required. If there are competing priorities of learning objectives, then the engagement with conflict can be deferred.

## Role of the educator

The curriculum alone cannot bring the desired social-emotional competencies. The role of educator is paramount in creating a safe space and stimulating experiences for students to trust, engage effectively, stretch, and explore.

The children believed they enhanced creativity and were exposed to new perspectives through the performance arts class. The educator introduced them to concepts, guided them through the activities and process using prompts, feedback, and fresh perspectives.

*"Even if we know something already, she brings new ways of looking at it, encouraging us to see things differently."*

The educator envisaged her role beyond teaching. She aimed to create free spaces within the structured sessions to enable the children to 'play' around and choose their engagement. The key functions of this role are to provide a reliable structure of learning that is reassuring to the students, design opportunities for exploration, articulation, and expression with regard to performance and conflict resolution, enable the learner to see connections with the real world and life around them, facilitate the transition of information into knowledge, continuously observe and make frameworks for learning simultaneously. It is important for the educator to know when to step in and when to step away in the learning process. They must give attention to each student, building a one-on-one relationship with them and creating a space for equal dialogue and communication.

The relationship of power between the educator and learner exists and cannot be removed completely. The educator is in charge of a session, which enables them to guide the students and facilitate learning. It, thus, requires accountability and responsibility. Authority should be exercised in an ethical way only where needed. If a student challenges the educator, she responds with curiosity by engaging the child through listening, processing, and reflecting, evoking trust, and prompting the child to make decisions as per their need. Assertion of power can never be used for punishment, humiliation, shaming, and labelling.

## Conclusion

I have attempted to highlight the interconnections between art practice and social emotional learning hoping to bring forth the vast potential of art-integrated pedagogy. Educators are encouraged to use the frameworks shared in this article to design sessions to enhance the social-emotional competencies of learners. However, I also believe that this pedagogy can be applied to varied contexts and should, therefore, be explored and studied extensively by different practitioners.

The educators and practitioners of other disciplines who are working with children, youth, and adolescents in areas of education, mental health, community development, etc., can apply different performing arts based exercises designed to their objectives with intentionality and create immersive and embodied experiences for the desired learning outcomes.

***Note:*** This article represents my immersive learning experience which has been supported generously by many people. I would like to thank Shabari Rao, artist and academic, who has designed the performing arts curriculum and facilitated this journey mentoring me every step of the way. My sincere gratitude to Debosmita, the educator and her lovely students for sharing their valuable experiences with me, and Savitha, co-founder and director, and all adults from Pramiti who welcomed me as a co-learner in this journey.

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# Revisiting a

Neeraja Raghavan

As a teacher, how often have you felt that you were knocking your head against a wall?

That no matter how hard you try, this student is simply *not getting it*?

And have you given up sometimes, realizing that you are human and can't possibly reach out to *every single child*?

If you have, then this account will resonate with you.

I recently spoke to a teacher with four decades of experience and was amazed at her description of the turning point in her teaching journey. It is a story with a familiar beginning: of a lady who slipped into the profession by chance and not choice.

Somewhere along the way, she came across a student who posed a huge challenge: he just didn't seem to absorb anything that she taught him. Try as she might, she couldn't succeed in getting across to the boy. He failed his end-of-year exams twice.

Frustrated with her persistent efforts falling flat each time, she went with the problem to her principal. At that time, the school policy allowed a teacher to request that the school dismiss a student who failed twice in the same grade. (When she spoke to me, however, this was no longer true.)

So this despairing teacher made a sound case before her principal to dismiss this 'good-for-nothing' boy, and answered every question posed by the principal with convincing answers to support her case. "Surely he was good at something?" insisted the principal. No, he wasn't, asserted this teacher firmly.

And so the errant boy was indeed dismissed.

The academic year ended, and the teacher resumed her teaching duties the next year. Seeing her batch of promoted students, she noticed the absence of that boy. Upon enquiring about his whereabouts, she received the shocking reply: "He is working in a cycle repair shop."

Photos courtesy: Aloka Mathur

# premise

That hit her like a ton of bricks.

Slowly, she came to know that he was the son of a gardener, who (what is worse) was a drunkard.

And she had ensured that such a boy never received a school education! Worse, she had even felt proud of that move!

This was the turning point in her life.

She switched instantly from being a teacher to a researcher.

What was the cause of a child not performing well?

Was it always sheer indifference?

Could there be other reasons?

With an undying thirst, this teacher began to learn about learning disabilities and methods of teaching children who faced difficulty. This was in the era before the internet, so there was no instant click of a button to help her. Her hunger to learn was the 'mouse' that she kept clicking. It was also the period when an awareness of children with special needs was low: unlike today.

As she began examining the specific difficulties that children faced, her paths crossed specialists in NIMHANS, Bengaluru, resource centres and workshops that she scoured the daily newspapers for announcements of. This led to her forming a section in her school for children with specific learning needs, where she and her equally enthusiastic colleague designed tailor-made lessons for their students.

As she developed customized methods to tackle specific learning difficulties of students, that boy she had been instrumental in dismissing kept popping up in her mind. And guess what she did? She went on a search: *for that student*! Not only did she find him two years later, but he was re-enrolled in her school and she succeeded in helping him pass the class X board exam!

Not surprisingly, this teacher won the NCERT Award for being an innovative teacher: but more than any award, her unique *retracing of steps to correct a biased premise* is priceless indeed.

What could well have continued as a mechanical emphasis on performance alone, heedless of underlying causes for failure, made this teacher go out of her way to reflect upon (and understand) the root cause of such difficulties.

Watch her detailed narration of her journey here: https://www.youtube.com/watch?v=thW_fh5sSQ0&t=9s

The author is founder-director of Thinking Teacher (www.thinkingteacher.in), an organization that networks with teachers across the country. Thinking Teacher aims to awaken and nurture the reflective practitioner within each teacher. Thinking Teacher's goal is to help build deep inquiry and rich learning into the teaching process. The author can be reached at <neeraja@thinkingteacher.in>.

# Terrarium: a fascinating

Shalini Sabikhi

Photos courtesy: Shalini Sabikhi

A gift-wrapped terrarium

Terrariums offer a low-maintenance, space-efficient, and visually appealing way to bring the beauty of Nature into our home or office. It is a mini garden or an ecosystem in a jar.

**Benefits of terrariums**

- Low maintenance: The closed environment helps retain moisture, reducing the need for frequent watering.
- Space-saving: Terrariums are compact and can be placed in small spaces like desktops, shelves, or window sills making them great options for people with limited space.
- Improved air quality: The plants in a terrarium release oxygen and help purify air, improving air quality in the surrounding environment.
- Educational: Terrariums allow you to observe the life cycle of plants and the interactions between different organisms up close. This makes them great educational tools for children.
- Aesthetic appeal: Terrariums can be visually appealing with the plants and decorative elements creating a miniature, self-contained ecosystem.
- Stress relief: Watching the plants in a terrarium grow and thrive can have a calming and therapeutic effect, helping reduce stress.
- Versatility: They can be designed in a variety of styles, from simple and minimalist to elaborate and artistic, making them suitable for different tastes, décor styles, and budget.
- A great gifting option: Terrariums are great gifts to people of any age on any occasion.

**How terrariums benefit students in their learning**

- Ecosystem: Students get a hands-on experience of the ecosystem, bio-diversity, photosynthesis, and the relationship between plants, soil, and water. In a closed terrarium they can also observe and learn about the water cycle. When water inside the closed terrarium heats up or when leaves of the plants give out water by transpiration, this water evaporates and deposits as moisture on the sides of the glass jar. On cooling, condensation takes place and the moisture gets converted into water droplets. Precipitation then occurs and the water droplets fall down on the plants.
- Plant care: They learn about plant growth and also how to care for the plants. This develops a sense of responsibility towards Nature.
- Scientific inquiry: Students can design experiments to learn more about plant growth and environmental factors.
- Environmental knowledge: In a terrarium, students deal with a miniature ecosystem. Having and maintaining terrariums can lead to conversations about conservation, habitat

# learning tool

destruction, and the impact of human activities on the environment.

- Get creative and artistic: Students can get creative with designing and decorating their terrariums. They can use a variety of plants, pebbles, shells, miniatures, etc., to decorate. The decoration in their terrariums can also be based on particular themes like jungle, village, urban jungle, temple, air force base, etc.
- Overall, terrariums help in multidisciplinary learning – science, ecology, and art.

**Requirements**

- Transparent glass/plastic jar
- Pebbles and marble chips
- Soil
- Moss
- Charcoal
- Brick powder
- Accessories (miniature garden toys)
- Succulent or any small indoor plant
- Water spray bottle

These things are usually available in plant nurseries or online.

**Method**

- Take a transparent glass/plastic jar of any shape and size.
- Wash with lukewarm water and dry it completely.
- Put a layer of white marble chips at the bottom. Excess water gets drained here.
- Keep a piece of cloth or polythene sheet over the marble chips (with small holes in it to allow water to flow down). This sheet will keep the coal powder from mixing with the marble chips.
- Top it with a layer of coal powder. This helps absorb any sitting water and prevents odour and bacteria build-up.
- Put a layer of brick powder over it.
- Above it add a layer of moss. This will prevent and stop the soil from slipping to the bottom during watering.
- Top that with compost and soil.
- Add accessories.

**Taking care of your terrarium**

- When lifting your glass terrarium, lift with both hands giving support at the bottom.
- Keep it in a spot with plenty of natural light (but not direct sunlight).
- Spray your terrarium with water every week or when the soil gets dry.
- Do not overwater.

The author is a social studies educator in Ahmedabad. She is interested in reading, travelling, gardening, art and craft, learning and sharing her learnings with others. She can be reached at <svs502@yahoo.com>.

NOTES FROM A TEACHER'S DIARY

# Nature is around us

Sana B Khan

Photo courtesy: Sana B Khan

It's in the chirping of birds amidst the blaring horns on the street, it's in the butterflies that find your home garden and make it their own, it's in the little peepal sapling growing through the cement wall.

When you go out for a walk, and if you really try to "see" it, you will always find Nature trying to connect with you. Sometimes, it's a leaf falling on your head and sometimes it's a bird that is flying by.

But what happens when you can't "see" the Nature around you?

It's simple, you see with your heart.

Working with students with visual impairment reminded me that what our eyes do for us is necessary, but if our hearts are closed, everything becomes blurry.

I always tell my students that the world around us – the birds, the trees, the insects – is always finding a way to reach us. All we need to do is to keep our hearts open to the experience.

For a person with visual impairment, the changes in the colours of trees, the discipline of ants as they go home, the bright colour of the jarul flowers might be difficult to see. But these things are not difficult to perceive.

As their teacher, it became important to me that my students also perceive the beautiful world around us. And so I began helping them build a connection with Nature. I remember the first time I brought a dead moth into our classroom. Before I could finish saying, "See, this is a dead insect, let's explore it," of the 11 students only three remained. The fear that the word "keeda" brought on, made them run out of the class, and at such speed that it still makes me laugh.

Since then, we have been doing everything possible to bridge this gap between us and Nature. We climb the jarul trees, pick up leaves every season to explore the changes in that tree, hug the peepal tree to see its girth, and most importantly, we don't run away when dead insects are brought into the classroom.

A few weeks ago, my student came running with something in his hand, "Ma'am, dekho kya mila?" I am sure you will all agree that when students say these words, our heart skips a beat, because children can find anything. But this time, my dear student had found a butterfly. A bushbrown that came into his classroom, fluttered by his ear, and sat on his sweaty palms.

I was the happiest when I saw the butterfly resting in his palms; it reminded me of the 11 students who had run away a year ago, when they heard the word 'insect'. Today, they all sat down to discuss butterflies and moths.

Our eyes allow us to see the world, but only an open heart can perceive it.

***Note:*** I would like to thank 'The Victoria Memorial School for the Blind' for all their support in making learning fun and accessible.

The author is a special educator working with students with visual impairment. She can be reached at <sanadoesntmail@gmail.com>.

# The Values that Constitute our Nation

Prakash Iyer

**Anubhuti:** Today was the first civics class and I did a pretty good job of generating strong interest and respect for the Indian Constitution. I was worried this class would be like last year when it became drab and boring. Students felt they only had to memorize the articles and that the Constitution didn't have any relevance to their daily lives. But this time, I think most students could feel its importance in their daily life.

**Ira:** It is indeed a difficult subject to teach. How do you generate interest in students when the text per se is quite difficult to understand? You can't even create stories around it like we do in history.

**Anubhuti:** Oh, but that is not true. We can and in fact we should do precisely that – build stories. Except these would be real narratives, not fictitious. There are so many court cases involving the Constitution. They might become complex for 9th grade students though. So, I narrated the story of how the Constituent Assembly was formed, and how they drafted the Constitution. I screened the first episode of Samvidhan, the TV series created by Shyam Benegal and then highlighted some key aspects of the story.

Firstly, I pointed out how the document was not the product of a few chosen people. It was the creation

Image generated using AI

of 389 representatives of the Indian polity including the then princely states and union territories, which makes the Constitution "ours". It is not a sacred book written by a few thinkers that we must worship. Rather, the document represents the aspirations of all citizens of India and how they constituted themselves as India.

I took the linguistic route to emphasize the importance of the constitution. I spent a lot of time explaining the word 'constitution' both as a noun and a verb. The word 'constitution' means "the way that something is formed from different parts". You see, when the Preamble says we *constitute India into*, we are saying that we are creating a new concept of India, which is defined by sovereign, democratic, and republic. I ended today's class with the Preamble.

Going forward, I plan to highlight the process of drafting the Constitution, the numerous steps each article went through before being finalized. It was not merely some people in subcommittees drafting the articles and everyone voting for it. It involved prolonged debates and even disagreements. Each episode of Samvidhan focuses on fundamental rights and disagreements between the members of the Constituent Assembly. How they had to debate, negotiate, and even give in when the final voting in subcommittees took place. After that the articles were presented in the parliament, where further dialogue happened.

The process of drafting the Constitution itself was democratic, which is why it is fair to say that in 1950 we *constituted* ourselves as a democratic republic and which is why we are who we are today in 2025.

**Ira:** But this is disconcerting. You mean there were disagreements among the members of the Constituent Assembly, and even then the articles were considered final and sacred? And generations later, we citizens must accept the Constitution the way it is?

**Anubhuti:** I disagree with that. We should respect the Constitution in spirit, but it does not help to see it as being sacred. The word 'sacred' implies that it has to be completely accepted as it is, and cannot be changed or tampered with. Of course, the Constitution needs to be interpreted and applied to reality, but it is not untouchable. The Constitution is not merely a set of rules; each article is based on some principles of democracy. The foundation of the Constitution is democracy, which necessitates prolonged dialogue between citizens. Secondly, the Constitution is not written in stone. We add layers of meaning to the text, and it can and has been amended many times – following a rigorous due process.

**Ira:** One question that is bound to come up is – how come we keep changing the Constitution and its meaning? Wouldn't that be wrong? We need to accept the Constitution in spirit rather than mere laws and words that could change overnight. The students need to *feel* for the Constitution and not look at it merely as a book of rules that ought to be followed.

**Anubhuti:** That is right. It is a tricky question and is bound to come from the students. But my response fits into what we have discussed so far. The Constitution itself has some rules on how it can be amended, and what can and cannot be amended. It is a flexible document which allows to be changed with times, but not so flexible that anything can be changed. For example, none of the fundamental rights or duties can be tampered with. We can add more meaning to it – which is done by every Supreme Court judgment that interprets one or more articles of the Constitution.

**Ira:** It makes the Constitution like a palimpsest. A fixed text which has layers of meaning added to it with every Supreme Court judgment that interprets the articles in it.

**Anubhuti:** That metaphor is very useful. It also makes it possible for me to explain how some aspects of the Constitution cannot be changed. It is too early to explain the basic structure doctrine to 9th grade students, but this metaphor is useful to say that the text of the Constitution will seem unchanged, but the meanings keep evolving with time. I am sure they will understand if I say that when we are running a school, we will read article 19(g), "All citizens shall have the right to practice any profession or to carry any occupation or business", differently than if we were running a provision store.

**Ira:** Or selling phones (ha ha ha).A richer comparison is between understanding the Right to Freedom when we are running a school as against when we are participating in a Parent Teacher meeting as parents.

**Anubhuti:** This also opens the point of how the popular discourse on the Constitution is in fact a

dialogue between the polity and the text of the Constitution. I was talking to this lawyer friend who said the term used in legal circles is *living constitutionalism*.

**Ira:** What is that?

**Anubhuti:** She was saying that in simple terms it just means that the text of the Constitution should – must – be interpreted in connection with current realities, social needs, and contestations. For instance, in 2009, the Right to Free and Compulsory Education Act amounted to addition to Article 21. Education became a fundamental right from there on. Moreover the act mandated that private schools also reserve 25% of their seats for students from disadvantaged sections of society. For Pramati Educational Trust, this judgment limited the right to freedom to run the organization the way they wanted. The Supreme Court emphasized that this freedom carries less force than the right to free and compulsory education that children from disadvantaged communities have.

And why all schools need to pitch in to help those students who do not get good schooling. It is not the responsibility of the government alone; it never has been. All the Supreme Court judgment* did was to emphasize that private schools must share the government's responsibility and work towards 100% education for all. And to that end they would have to sacrifice part of their Right to Freedom to run an organization. This judgment has added layers to Right to Freedom, Right to Equality, and Cultural and Educational Rights.

The Supreme Court demonstrated living constitutionalism by saying that the State's right to impose reservations on private schools carries more force than Right to Freedom, because education is a public good and it is the private school's responsibility to fulfil its duties.

Basically, I pointed out how the Right to Free and Compulsory Education Act was a necessity in today's scenario.

*Judgment of the Pramati Educational and Cultural Trust vs. Union of India. https://indiankanoon.org/doc/32468867/

**Ira:** That is dense Anubhuti. You think you can explain this to our students? And do you think they will comprehend this?

**Anubhuti:** Yeah, you are right. I am explaining this to you, because this conversation changed my view towards the Constitution. That changed view will transform the way I present the Constitution to students. That we have the right to change it does not mean we can do what we want. We must be justified and explain how the new interpretation is related to new social realities. What has happened today that was not there until yesterday, and it demands that we suggest that the Constitution be changed?

In short, I would say our duty towards the Constitution – and towards the concept of a democratic republic which India is – takes precedence over our right to demand change.

You see this is the beauty of philosophical thinking. When our view towards something has changed, it changes the way we talk about it. Then we automatically use different and right-er words than we did until now. We teachers need to know more about the Constitution (obviously), but also know how we are thinking about it. We may not be teaching all of this in class, at least not in 9th grade, but that doesn't mean this change in our way of thinking is not needed. Or this debate that you and I are having. That is equally, if not more, important.

**Ira:** Yeah, a friend once told me the definition of philosophy he likes most is – thinking about thinking. You are saying precisely that. We teach the Constitution in simple terms to 9th class students, but our deeper understanding of the Constitution will and should inform the way we teach it. Hopefully, the class will learn how to think philosophically without knowing that they are.

The author teaches Philosophy of Education and Curriculum theory at Azim Premji University. He can be reached at <prakash.iyer@apu.edu.in>.

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